

Chapter 1 Fourier series

1.1 Appearance of Fourier series

The birth of Fourier series can be traced back to the solutions of wave equation in the work of Bernoulli and the heat equation in the work of Fourier.

Consider an elastic string of finite length l fixed at the end points $x = 0$ and $x = l$. At time, say $t = 0$, it is distorted from the equilibrium position and allowed to vibrate. The problem is to find the vibrations of the string at any point x and any time $t > 0$. The vibration of the string is governed by the linear partial differential equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}, \quad u(x, 0) = f(x), \quad \frac{\partial}{\partial t} u(x, 0) = g(x). \quad (1.1)$$

Here c^2 denotes a physical constant, $f(x)$ gives the initial position and $g(x)$ gives the initial velocity.

In 1747, D. Alembert obtained the solution of (1.1) in the form

$$u(x, t) = \frac{1}{2} [f(x + ct) + f(x - ct)] + \frac{1}{2c} \int_{x-ct}^{x+ct} g(y) dy.$$

However, in 1753, Bernoulli had a different idea of solving this equation which was based on the observation that the functions $\sin \lambda ct$, $\sin \lambda x$, and $\cos \lambda ct$, $\cos \lambda x$ satisfy the equation for any $\lambda \in \mathbb{R}$. If we choose $\lambda = \frac{n\pi}{l}$, $n = 1, 2, 3, \dots$ they also satisfy the

boundary conditions. As the equation is linear, he argued that any superposition of such solutions will also be a solution.

To explain this method further, let us use the method of separation of variables. Let $u(x, t) = F(x)G(t)$. Substituting for u in (1.1), we get $c^2 F''(x)G(t) = F(x)G''(t)$, which can be rewritten as

$$\frac{F''(x)}{F(x)} = \frac{1}{c^2} \frac{G''(t)}{G(t)}. \quad (1.2)$$

In (1.2), the left hand side is a function of x and the right hand side is a function of t . This is possible only if

$$\frac{F''(x)}{F(x)} = k, \quad \frac{1}{c^2} \frac{G''(t)}{G(t)} = k,$$

where k is a constant. Thus we end up with two differential equations,

$$F''(x) - kF(x) = 0, \quad (1.3)$$

$$G''(t) - kc^2 G(t) = 0. \quad (1.4)$$

Since the string is fixed at end points $x = 0$ and $x = l$, we have the boundary conditions

$$u(0, t) = 0; u(l, t) = 0, \quad t > 0, \quad (1.5)$$

which are satisfied if we assume

$$F(0) = F(l) = 0. \quad (1.6)$$

The vibrations of the string at time t depend upon the initial deflection and initial velocity. Let f denote the initial deflection and g denote the initial velocity. Thus the initial conditions are given by

$$u(x, 0) = f(x), \quad (1.7)$$

$$\frac{\partial u}{\partial t}(x, 0) = g(x). \quad (1.8)$$

Notice that the constant k which appears in (1.3) and (1.4) is a real number. Hence it can be positive or negative or zero.

Case 1. If $k = 0$, then (1.3) and (1.4) become $F''(x) = 0$; $G''(t) = 0$. Solving for F , we get $F(x) = Ax + B$, where A and B are constants. The boundary conditions force $A = B = 0$ and hence $F = 0$.

Case 2. Let $k = \mu^2$ be positive. Substituting in (1.3), we get $F''(x) - \mu^2 F(x) = 0$. In this case, the solution is given by $F(x) = Ae^{\mu x} + Be^{-\mu x}$, where A and B are constants. Once again the boundary conditions force F to be zero.

Case 3. Let $k = -\lambda^2$. Then (1.3) becomes $F''(x) + \lambda^2 F(x) = 0$. On solving, we get $F(x) = A \cos \lambda x + B \sin \lambda x$, where A and B are constants. Applying $F(0) = 0$, we get $A = 0$.

Thus, $F(x) = B \sin \lambda x$. The equation $F(l) = 0$ leads to $B \sin \lambda l = 0$. Then, we get either $B = 0$ or $\sin \lambda l = 0$. In order to get a non-trivial solution, we assume that $B \neq 0$. Thus, $\sin \lambda l = 0$, which leads to $\lambda l = n\pi, n \in \mathbb{Z}$. As $\sin 0 = 0$ and $\sin(-n) = -\sin n$, it is enough to take $n \in \mathbb{N}$. Thus we have $\lambda_n = \frac{n\pi}{l}, n = 1, 2, 3, \dots$ and

$$F_n(x) = B_n \sin \lambda_n x.$$

Now (1.4) becomes $G_n''(t) + \lambda_n^2 c^2 G_n(t) = 0$. On solving we get,

$$G_n(t) = C_n \cos(c\lambda_n t) + D_n \sin(c\lambda_n t).$$

Let $u_n(x, t) = B_n \sin \lambda_n x (C_n \cos(c\lambda_n t) + D_n \sin(c\lambda_n t))$. It is clear that for each n , $u_n(x, t)$ satisfies the wave equation with the correct boundary conditions.

The function u_n also satisfies the initial conditions

$$u_n(x, 0) = B_n C_n \sin \lambda_n x, \quad \frac{\partial u_n}{\partial t}(x, 0) = c B_n D_n \lambda_n \sin \lambda_n x.$$

Consequently, the superpositions $u = \sum_{n=1}^{\infty} u_n$ solves the wave equation with the boundary conditions $u(0, t) = u(l, t) = 0$. However, the initial conditions will be satisfied only if

$$f(x) = \sum_{n=1}^{\infty} B_n C_n \sin \lambda_n x,$$

and $g(x) = c \sum_{n=1}^{\infty} B_n D_n \lambda_n \sin \lambda_n x$. At this point, it is not clear whether any f or g can be expanded in terms of $\sin \lambda_n x$ as above and hence the solution of the wave equation obtained by Bernoulli was incomplete. In 1807, Fourier was working on the initial value problem for the heat equation

$$\frac{\partial u}{\partial t}(x, t) = \frac{\partial^2 u}{\partial x^2}(x, t), \quad u(x, 0) = f(x).$$

Let us assume the same boundary conditions as above, namely $u(0, t) = u(l, t) = 0$ which means the temperature at the end points is kept at zero throughout. Proceeding as in the case of wave equation, we get the elementary solutions

$$u_n(x, t) = A_n (\sin \lambda_n x) e^{-\lambda_n^2 t}.$$

The superposition $u(x, t) = \sum_{n=1}^{\infty} u_n(x, t)$, which solves the heat equation, satisfies the initial condition provided

$$f(x) = \sum_{n=1}^{\infty} A_n \sin \lambda_n x.$$

Fourier asserted that any f satisfying $f(0) = f(l) = 0$ can be expanded in terms of $\sin \lambda_n x$ as above.

1.2 The formal Fourier series and some special kernels

In Section 1.1, we mentioned the idea of expanding a function $f(x)$ satisfying $f(0) = f(l) = 0$ in terms of $\sin \frac{n\pi}{l}x$, $n = 1, 2, \dots$. Note that the functions $\sin \frac{n\pi}{l}x$ are periodic with period $2l$ and hence f also has to be periodic in order to be expanded as above.

Given f with $f(0) = f(l) = 0$, we can always consider it as a periodic function of period $2l$ defined on the whole of $\mathbb{R}$ and hence it is reasonable to expect an expansion of the above type.

For the sake of simplicity, we take $l = \pi$ and consider functions f that are 2π -periodic. Rather than restricting our attention to functions satisfying $f(0) = f(\pi) = 0$, we consider all 2π periodic functions. In this case, besides $\sin nx$, we also need to include $\cos nx$ in the expansion of the type

$$f(x) = \sum_{n=0}^{\infty} (a_n \cos nx + b_n \sin nx).$$

Using the formulae, $\cos nx = \frac{1}{2}(e^{inx} + e^{-inx})$, $\sin nx = \frac{1}{2i}(e^{inx} - e^{-inx})$, we can rewrite the above expansion as

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx}.$$

This is a reasonable expansion as f is 2π -periodic and so is each e^{inx} . The family of functions e^{inx} have the interesting property, which we call orthogonality,

$$\int_{-\pi}^{\pi} e^{inx} e^{-imx} dx = 0 \text{ or } 2\pi,$$

depending on whether $n \neq m$ or $n = m$. Therefore, if termwise integration is allowed, then the expansion

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx},$$

leads us to

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx = c_n.$$

This formula, due to Fourier, allows us to calculate c_n in the expansion. We can now formally introduce the Fourier series.

Definition 1.2.1. Let f be a Lebesgue integrable function on $[-\pi, \pi]$. Then the n^{th} Fourier coefficient of f is defined by

$$\widehat{f}(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx, \quad n \in \mathbb{Z}.$$

The Fourier series of f is formally defined by $f(x) \sim \sum_{n=-\infty}^{\infty} \widehat{f}(n) e^{inx}$.

Example 1.2.2. Let $f(x) = x$, $-\pi \leq x \leq \pi$. Then, the Fourier coefficients of f are obtained as follows: for $n \neq 0$

$$\widehat{f}(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} x e^{-inx} dx$$

$$\begin{aligned}
&= \frac{1}{2\pi} \int_{-\pi}^{\pi} x d\left(\frac{e^{-inx}}{-in}\right) \\
&= \frac{-1}{2\pi in} \int_{-\pi}^{\pi} x d(e^{-inx}) \\
&= \frac{-1}{2\pi in} \left\{ x e^{-inx} \Big|_{-\pi}^{\pi} - \int_{-\pi}^{\pi} e^{-inx} dx \right\} \\
&= \frac{-1}{2\pi in} \{2\pi(-1)^n - 0\}, \quad n \neq 0.
\end{aligned}$$

Thus $\hat{f}(n) = \frac{(-1)^{n+1}}{in}$, $n \neq 0$. But $\hat{f}(0) = \frac{1}{2\pi} \int_{-\pi}^{\pi} x dx = 0$. In this case, the Fourier series of f can be formally written as

$$f(x) \sim \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} \frac{(-1)^{n+1}}{in} e^{inx} = 2 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin nx.$$

Example 1.2.3. Let $f(x) = x^2$, $-\pi \leq x \leq \pi$. Then the n^{th} Fourier coefficient of f is given by

$$\begin{aligned}
\hat{f}(n) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} x^2 e^{-inx} dx \\
&= \frac{1}{2\pi} \int_{-\pi}^{\pi} x^2 d\left(\frac{e^{-inx}}{-in}\right).
\end{aligned}$$

Using integration by parts, we get,

$$\hat{f}(n) = \frac{2(-1)^n}{n^2}, \quad n \neq 0.$$

But

$$\hat{f}(0) = \frac{1}{2\pi} \int_{-\pi}^{\pi} x^2 dx = \frac{\pi^2}{3}.$$

In this case, the formal Fourier series of f can be written as

$$\begin{aligned} f(x) &\sim \frac{\pi^2}{3} + \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} \frac{2(-1)^n}{n^2} e^{inx} \\ &= \frac{\pi^2}{3} + \sum_{n=1}^{\infty} \frac{4(-1)^n}{n^2} \cos nx. \end{aligned}$$

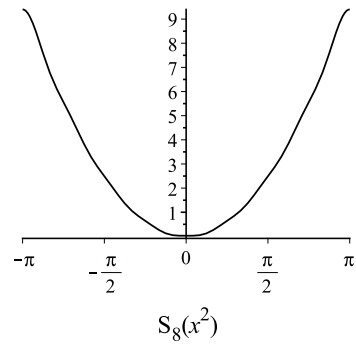
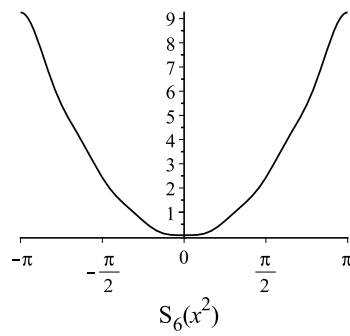
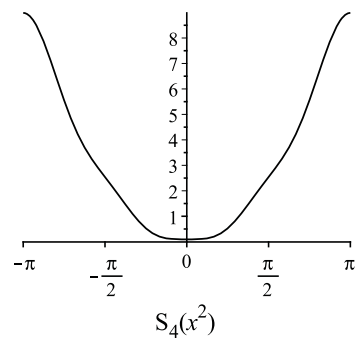
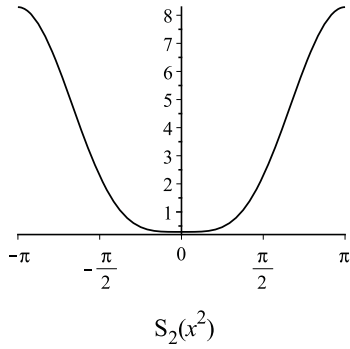


Figure 1.1: Fourier series of x^2 .

Consider the Fourier series

$$\begin{aligned}
f(x) &\sim \sum_{n=-\infty}^{\infty} \widehat{f}(n) e^{inx} \\
&= \widehat{f}(0) + \sum_{n=1}^{\infty} \widehat{f}(-n) e^{-inx} + \sum_{n=1}^{\infty} \widehat{f}(n) e^{inx} \\
&= \widehat{f}(0) + \sum_{n=1}^{\infty} [\widehat{f}(-n) + \widehat{f}(n)] \cos nx \\
&\quad + i \sum_{n=1}^{\infty} [\widehat{f}(n) - \widehat{f}(-n)] \sin nx,
\end{aligned}$$

using $e^{\pm inx} = \cos nx \pm i \sin nx$. If f is an even function, then $\widehat{f}$ will also be an even function. In fact,

$$\begin{aligned}
\widehat{f}(-n) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{inx} dx \\
&= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(-x) e^{inx} dx \\
&= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(-x) e^{-(-inx)} dx \\
&= \widehat{f}(n),
\end{aligned}$$

by applying a change of variable $x = -y$. In this case, the resulting Fourier series becomes $f(x) \sim \widehat{f}(0) + 2 \sum_{n=1}^{\infty} \widehat{f}(n) \cos nx$. On the other hand, if f is an odd function, then $\widehat{f}$ will also be an odd function and the Fourier series takes the form

$$f(x) \sim 2i \sum_{n=1}^{\infty} \widehat{f}(n) \sin nx.$$

1.2.1 Dirichlet kernel

The formal Fourier series will represent the function f provided the partial sums of the series converges to f . We take the symmetric partial sum

$$S_N f(x) = \sum_{n=-N}^N \widehat{f}(n) e^{inx},$$

which can be written in the form

$$S_N f(x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(y) D_N(x-y) dy,$$

where the function D_N , called the Dirichlet kernel, is given by

$$D_N(x) = \sum_{n=-N}^N e^{inx}.$$

Indeed, from the definition,

$$\widehat{f}(n) e^{inx} = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(y) e^{-in(y-x)} dy,$$

and hence

$$\sum_{n=-N}^N \widehat{f}(n) e^{inx} = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(y) \left(\sum_{n=-N}^N e^{-in(y-x)} \right) dy.$$

The kernel D_N can be explicitly calculated:

$$D_N(x) = \frac{\sin(2N+1)\frac{x}{2}}{\sin(\frac{x}{2})}. \tag{1.9}$$

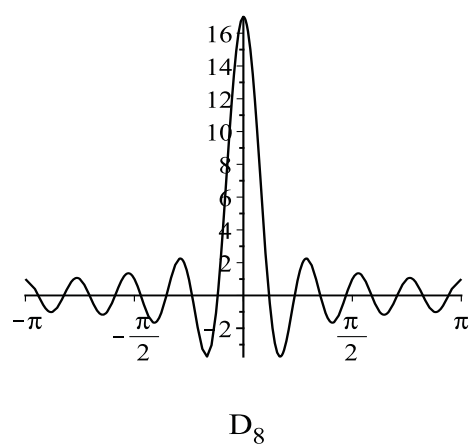
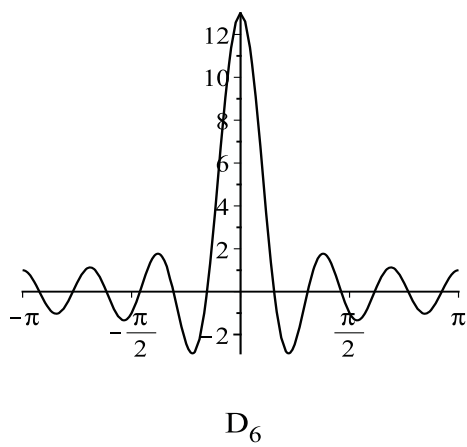
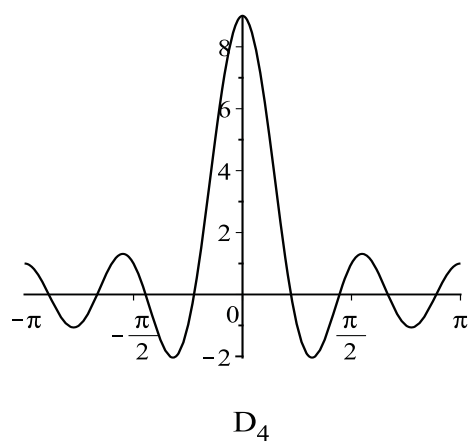
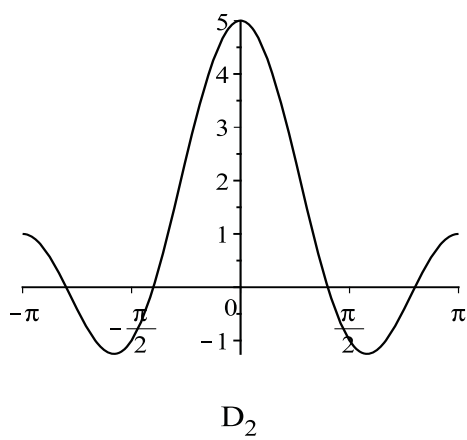


Figure 1.2: Dirichlet kernel.

In order to establish (1.9), consider

$$\begin{aligned}
D_N(x) &= \sum_{n=-N}^N e^{inx} \\
&= \sum_{n=-N}^{-1} e^{inx} + \sum_{n=1}^N e^{inx} + 1 \\
&= -1 + \sum_{n=0}^N e^{-inx} + \sum_{n=0}^N e^{inx} \\
&= -1 + \sum_{n=0}^N (e^{-ix})^n + \sum_{n=0}^N (e^{ix})^n.
\end{aligned}$$

Writing $w = e^{ix}$, we have

$$\begin{aligned}
D_N(x) &= -1 + \sum_{n=0}^N w^{-n} + \sum_{n=0}^N w^n \\
&= -1 + \frac{1 - w^{-(N+1)}}{1 - w^{-1}} + \frac{1 - w^{(N+1)}}{1 - w} \\
&= -1 + \frac{w - w^{-N}}{w - 1} + \frac{1 - w^{(N+1)}}{1 - w} \\
&= \frac{w^{-N} - w^{N+1}}{1 - w} \\
&= \frac{w^{-1/2}(w^{-N} - w^{N+1})}{w^{-1/2}(1 - w)} \\
&= \frac{w^{-N-1/2} - w^{N+1/2}}{w^{-1/2} - w^{1/2}} \\
&= \frac{\sin(2N+1)\frac{x}{2}}{\sin(\frac{x}{2})}.
\end{aligned}$$

Observe that, from the very definition, the Fourier coefficients of D_N are given by

$$\widehat{D}_N(n) = \begin{cases} 1 & \text{if } |n| \leq N \\ 0 & \text{otherwise.} \end{cases}$$

1.2.2 Dirichlet problem for the disc and Poisson kernel

Let $\mathbb{D}$ denote the unit disc in the complex plane i.e. $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$. Given a continuous function f on the boundary of $\mathbb{D}$ we are interested in finding a function u on $\mathbb{D}$ satisfying the Laplace equation $\Delta u = 0$ with the boundary condition $\lim_{r \rightarrow 1} u(re^{i\theta}) = f(\theta)$. This is called the Dirichlet problem for the unit disc $\mathbb{D}$. The Laplace equation in polar coordinates takes the form

$$\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} = 0.$$

Integrating the equation against $e^{-in\theta} d\theta$ we see that

$$r^2 \frac{\partial^2}{\partial r^2} u_n(r) + r \frac{\partial}{\partial r} u_n(r) + \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{\partial^2 u}{\partial \theta^2}(r, \theta) e^{-in\theta} d\theta = 0,$$

where

$$u_n(r) = \frac{1}{2\pi} \int_{-\pi}^{\pi} u(r, \theta) e^{-in\theta} d\theta.$$

Integrating by parts, the third term becomes $-n^2 u_n(r)$ and hence we get

$$r^2 \frac{\partial^2}{\partial r^2} u_n(r) + r \frac{\partial}{\partial r} u_n(r) - n^2 u_n(r) = 0.$$

Clearly, $u_n(r) = a_n r^{|n|}$ satisfies the above equation and the initial conditions $\lim_{r \rightarrow 1} u(r, \theta) = f(\theta)$ leads to $\lim_{r \rightarrow 1} u_n(r) = \widehat{f}(n)$. Hence, $a_n = \widehat{f}(n)$ and the solution $u(r, \theta)$ has the formal expansion

$$u(r, \theta) = \sum_{n=-\infty}^{\infty} \widehat{f}(n) r^{|n|} e^{in\theta}.$$

As before, we can write this in a compact form. Let us define a kernel $P_r(\theta)$, called the Poisson kernel, by

$$P_r(\theta) = \sum_{n=-\infty}^{\infty} r^{|n|} e^{in\theta}.$$

Then $u(r, \theta)$ takes the form

$$u(r, \theta) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(\varphi) P_r(\theta - \varphi) d\varphi.$$

The kernel $P_r(\theta)$ can also be calculated explicitly. Indeed, we have

$$\begin{aligned} P_r(\theta) &= \sum_{n=-\infty}^{\infty} r^{|n|} e^{in\theta} \\ &= 1 + \sum_{n=1}^{\infty} r^n e^{in\theta} + \sum_{n=1}^{\infty} r^n e^{-in\theta} \\ &= 1 + re^{i\theta}(1 - re^{i\theta})^{-1} + re^{-i\theta}(1 - re^{-i\theta})^{-1} \\ &= 1 + \frac{2r \cos \theta - 2r^2}{1 - 2r \cos \theta + r^2} \\ &= \frac{1 - r^2}{1 - 2r \cos \theta + r^2}. \end{aligned}$$

It can be easily shown that the Fourier coefficients of P_r are given by

$$\widehat{P_r}(n) = r^{|n|}.$$

1.2.3 The heat kernel

Returning to the heat equation, let $u(x, t)$ solve the problem

$$\frac{\partial u(x, t)}{\partial t} = \frac{\partial^2 u(x, t)}{\partial x^2},$$

$$u(x, 0) = f(x), \quad -\pi \leq x \leq \pi.$$

For each n , let

$$u_n(t) = \frac{1}{2\pi} \int_{-\pi}^{\pi} u(x, t) e^{-inx} dx.$$

Then we have

$$\frac{d}{dt}(u_n(t)) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{\partial^2 u(x, t)}{\partial x^2} e^{-inx} dx,$$

which after integration by parts, leads to

$$\frac{d}{dt}(u_n(t)) = -n^2 u_n(t).$$

The solution of this equation is given by $u_n(t) = a_n e^{-tn^2}$. The initial condition $u(x, 0) = f(x)$ gives $a_n = \widehat{f}(n)$ and hence,

$$u(x, t) = \sum_{n=-\infty}^{\infty} \widehat{f}(n) e^{-tn^2} e^{inx}.$$

As before, we can rewrite this as

$$u(x, t) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(y) h_t(x - y) dy,$$

where $h_t(x)$, called the heat kernel, is given by

$$h_t(x) = \sum_{n=-\infty}^{\infty} e^{-tn^2} e^{inx}.$$

Unlike the Poisson or Dirichlet kernel, we do not have a closed form expression for $h_t(x)$.

1.3 Uniqueness of Fourier series and some consequences

Let $\mathbb{T}$ denote the unit circle. A function on $\mathbb{T}$ can be treated as a function on $\mathbb{R}$ which is 2π -periodic. Let $\mathcal{C}(\mathbb{T})$ denote collection of all continuous functions on $\mathbb{T}$. The following theorem gives the uniqueness of Fourier series.

Theorem 1.3.1. *Let $f \in \mathcal{C}(\mathbb{T})$. Suppose that $\widehat{f}(n) = 0 \forall n \in \mathbb{Z}$ then $f = 0$.*

Proof. For $0 < \delta < \pi$, to be fixed soon, consider the function $p_\delta(x) = (1 + \cos x - \cos \delta)$. Then we claim that $p_\delta(x) \geq 1$ on $[-\delta, \delta]$ and $p_\delta(x) < 1$ on $[-\delta, \delta]^c$. Since $\cos x$ is an increasing function on $[-\pi, 0]$ and a decreasing function on $[0, \pi]$ we have for $x \in [-\delta, 0]$, $\cos x - \cos \delta \geq 0$. Then $p_\delta(x) = 1 + \cos x - \cos \delta \geq 1$ on $[-\delta, 0]$.

On the other hand, if $x \in [0, \delta]$ then $\cos x \geq \cos \delta$ and hence $p_\delta(x) \geq 1$ on $[0, \delta]$. However, if $-\pi < x < -\delta$ then $\cos x < \cos(-\delta) = \cos \delta$.

Hence, $p_\delta(x) < 1$ on $[-\pi, -\delta]$. Similarly, if $\delta < x < \pi$, then $\cos x < \cos \delta$ which leads to $p_\delta < 1$ on $[\delta, \pi]$.

Let $f = g + ih$, where g and h are real valued functions on $\mathbb{T}$. We prove that $f(x) = 0$ at every x . Without loss of generality, we take $x = 0$. Suppose $f(0) \neq 0$. Then, again without loss of generality, we can assume that $g(0) > 0$ and we will arrive at a contradiction using the continuity of g .

Choose $\delta > 0$ such that $g(x) > \frac{1}{2}g(0)$ on $[-\delta, \delta]$. Define p_δ with this δ and take $P_n(x) = p_\delta(x)^n$. By the hypothesis,

$$\int_{-\pi}^{\pi} f(x) P_n(x) dx = 0,$$

which gives

$$\int_{-\pi}^{\pi} g(x) P_n(x) dx = 0.$$

On the other hand, $\int_{[-\delta, \delta]^c} g(x) P_n(x) dx$ goes to 0 by dominated convergence theorem since $p_\delta(x) < 1$. This shows that

$$\lim_{n \rightarrow \infty} \int_{-\delta}^{\delta} g(x) P_n(x) dx = 0.$$

On the other hand, $P_n(x) \geq 1$ on this interval and hence,

$$\int_{-\delta}^{\delta} g(x) P_n(x) dx \geq \delta g(0).$$

This contradiction proves our claim that $f(0) = 0$. □

Corollary 1.3.2. *Suppose $f \in L^1(\mathbb{T})$ and $\widehat{f}(n) = 0 \forall n \in \mathbb{Z}$. Then $f(x) = 0$ a. e.*

Proof. Let $f \in L^1(\mathbb{T})$. Define $g(x) := \int_{-\pi}^x f(t) dt$, $-\pi \leq x \leq \pi$, then g is continuous and $g' = f$ a.e. Further, for all $n \neq 0$, $\widehat{g}(n) = \frac{i\widehat{f}}{n}$ so that $\widehat{g}(n) = 0$.

Applying Theorem 1.3.1 to the continuous function $g(x) - \widehat{g}(0)$, we conclude that $g(x) = \widehat{g}(0)$ is a constant. Hence, $f = 0$ a.e. □

Theorem 1.3.3. *Let $f \in \mathcal{C}(\mathbb{T})$ be such that $\widehat{f} \in l^1(\mathbb{Z})$. Then the Fourier series of f converges uniformly to f on $\mathbb{T}$.*

Proof. As $\widehat{f} \in l^1(\mathbb{Z})$, $\sum_{n=-\infty}^{\infty} |\widehat{f}(n)| < \infty$. Define $S_N f(x) = \sum_{n=-N}^N \widehat{f}(n) e^{inx}$. We claim that $S_N(f) \rightarrow f$ uniformly on $\mathbb{T}$.

For $N > M$, we have, for all $x \in \mathbb{T}$,

$$\begin{aligned}
|S_N f(x) - S_M f(x)| &= \left| \sum_{n=-N}^N \widehat{f}(n) e^{inx} - \sum_{n=-M}^M \widehat{f}(n) e^{inx} \right| \\
&= \left| \sum_{N \geq |n| > M} \widehat{f}(n) e^{inx} \right| \\
&\leq \sum_{N \geq |n| > M} |\widehat{f}(n)|.
\end{aligned}$$

Thus by applying Cauchy's criterion for uniform convergence, one can conclude that $S_N(f)$ converges to a continuous function g (uniformly) on $\mathbb{T}$. Further $\widehat{g}(n) = \widehat{f}(n) \forall n \in \mathbb{Z}$. In fact, for all n

$$\begin{aligned}
\widehat{g}(n) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} g(x) e^{-inx} dx \\
&= \frac{1}{2\pi} \int_{-\pi}^{\pi} \left(\sum_{m=-\infty}^{\infty} \widehat{f}(m) e^{imx} \right) e^{-inx} dx \\
&= \frac{1}{2\pi} \sum_{m=-\infty}^{\infty} \int_{-\pi}^{\pi} \widehat{f}(m) e^{i(m-n)x} dx \\
&= \widehat{f}(n).
\end{aligned}$$

Here, changing the order of integral and the sum was possible because the convergence is uniform. As $(f - g)\widehat{}(n) = 0 \forall n \in \mathbb{Z}$, we conclude that $f = g$. $\square$

Corollary 1.3.4. *If $f \in \mathcal{C}^2(\mathbb{T})$, then the Fourier series of f converges uniformly to f on $\mathbb{T}$.*

Proof. By the theorem, it is enough to prove that $\widehat{f} \in l^1(\mathbb{Z})$. We shall show that $\widehat{f}(n) = O\left(\frac{1}{|n|^2}\right)$ as $|n| \rightarrow \infty$. Consider, for $n \neq 0$,

$$\begin{aligned}\widehat{f}(n) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx \\ &= \frac{1}{2\pi} \left\{ f(x) \frac{e^{-inx}}{-in} \Big|_{-\pi}^{\pi} + \frac{1}{in} \int_{-\pi}^{\pi} f'(x) e^{-inx} dx \right\} \\ &= \frac{1}{2\pi in} \int_{-\pi}^{\pi} f'(x) e^{-inx} dx \\ &= \frac{1}{2\pi in} \left\{ f'(x) \frac{e^{-inx}}{-in} \Big|_{-\pi}^{\pi} + \frac{1}{in} \int_{-\pi}^{\pi} f''(x) e^{-inx} dx \right\}.\end{aligned}$$

Thus

$$\widehat{f}(n) = -\frac{1}{2\pi n^2} \int_{-\pi}^{\pi} f''(x) e^{-inx} dx.$$

Let $M = \max_{x \in [-\pi, \pi]} |f''(x)|$. Thus $|\widehat{f}(n)| \leq \frac{M}{n^2}$ for all $n \neq 0$. Consequently,

$$\begin{aligned}\sum_{n=-\infty}^{\infty} |\widehat{f}(n)| &\leq |\widehat{f}(0)| + \sum_{n \neq 0} |\widehat{f}(n)| \\ &\leq |\widehat{f}(0)| + 2M \sum_{n=1}^{\infty} \frac{1}{n^2} \\ &< \infty,\end{aligned}$$

which shows that $\widehat{f} \in l^1(\mathbb{Z})$. □

1.4 Convolution theory

Before defining convolution of two functions, we observe the following fact.

Proposition 1.4.1. *If f is a periodic function of period 2π , then*

$$\int_{-\pi}^{\pi} f(x+a)dx = \int_{-\pi}^{\pi} f(x)dx = \int_{-\pi+a}^{\pi+a} f(x)dx.$$

Proof. Consider $\int_{-\pi}^{\pi} f(x+a)dx$. Taking $x \mapsto x+a$, we get $\int_{-\pi}^{\pi} f(x+a)dx = \int_{-\pi+a}^{\pi+a} f(x)dx$.

If $a > 0$, then

$$\begin{aligned} \int_{-\pi+a}^{\pi+a} f(x)dx &= \int_{-\pi+a}^{\pi} f(x)dx + \int_{\pi}^{\pi+a} f(x)dx \\ &= \int_{-\pi}^{\pi} f(x)dx - \int_{-\pi}^{-\pi+a} f(x)dx + \int_{\pi}^{\pi+a} f(x)dx \\ &= \int_{-\pi}^{\pi} f(x)dx - \int_{-\pi}^{-\pi+a} f(x+2\pi)dx + \int_{\pi}^{\pi+a} f(x)dx \\ &= \int_{-\pi}^{\pi} f(x)dx - \int_{\pi}^{\pi+a} f(y)dy + \int_{\pi}^{\pi+a} f(x)dx = \int_{-\pi}^{\pi} f(x)dx. \end{aligned}$$

Similarly, we can prove the result when $a < 0$. □

Definition 1.4.2. For $f, g \in L^1(\mathbb{T})$, the convolution of f and g is defined as

$$(f * g)(x) := \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x-y) g(y) dy.$$

Immediately we observe that $f * g = g * f$. In fact,

$$(f * g)(x) = \frac{1}{2\pi} \int_{x-\pi}^{x+\pi} f(t)g(x-t)dt = \frac{1}{2\pi} \int_{-\pi}^{\pi} g(x-t)f(t)dt = (g * f)(x).$$

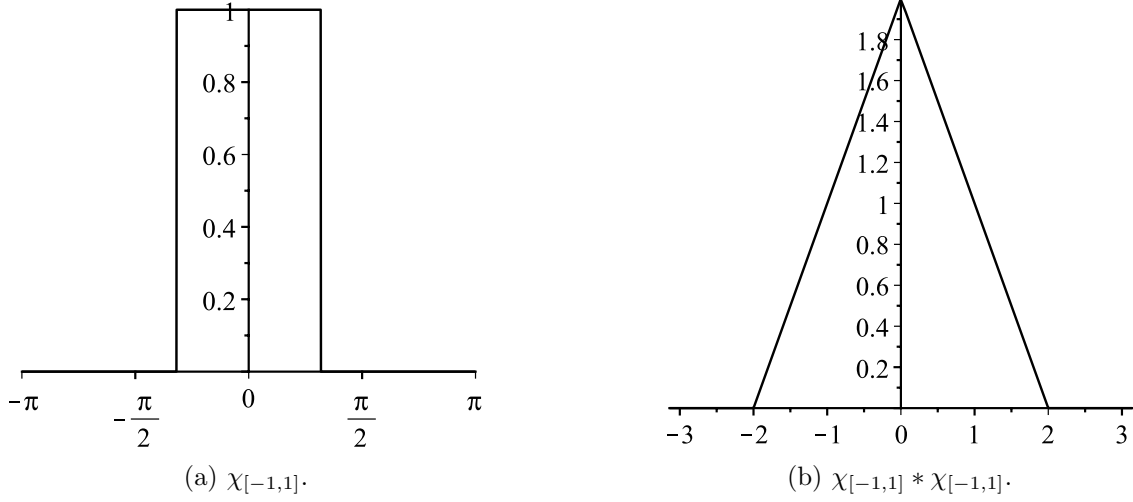


Figure 1.3

Theorem 1.4.3. $L^1(\mathbb{T})$ is a commutative Banach algebra.

Proof. We know that $L^1(\mathbb{T})$ is a Banach space. Let $f, g \in L^1(\mathbb{T})$. Consider

$$\begin{aligned} \|f * g\|_1 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} |f * g(x)| dx \\ &= \frac{1}{(2\pi)^2} \int_{-\pi}^{\pi} \left| \int_{-\pi}^{\pi} f(x-y) g(y) dy \right| dx \\ &\leq \frac{1}{(2\pi)^2} \int_{-\pi}^{\pi} \left(\int_{-\pi}^{\pi} |f(x-y)| |g(y)| dy \right) dx. \end{aligned}$$

Applying Fubini's theorem we get,

$$\begin{aligned}
\|f * g\|_1 &\leq \frac{1}{(2\pi)^2} \int_{-\pi}^{\pi} \left(\int_{-\pi}^{\pi} |f(x-y)| dx \right) |g(y)| dy \\
&= \frac{1}{2\pi} \int_{-\pi}^{\pi} \|f\|_1 |g(y)| dy \\
&= \|f\|_1 \|g\|_1 < \infty.
\end{aligned}$$

Thus $f * g \in L^1(\mathbb{T})$ and $\|f * g\|_1 \leq \|f\|_1 \|g\|_1$. We have already proved that $f * g = g * f$.

In order to prove associativity, consider, for $x \in \mathbb{R}$,

$$\begin{aligned}
(f * g) * h(x) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} (f * g)(y) h(x-y) dy \\
&= \frac{1}{2\pi} \int_{-\pi}^{\pi} \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) g(-t+y) dt \right) h(x-y) dy \\
&= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) \left(\frac{1}{2\pi} \int_{x-\pi}^{x+\pi} g(-t-u+x) h(u) du \right) dt \\
&= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) (g * h)(x-t) dt \\
&= [f * (g * h)](x).
\end{aligned}$$

By the linearity of integral, it follows that $(f + g) * h = (f * h) + (g * h)$ and $\alpha f * g = f * \alpha g$. Thus $L^1(\mathbb{T})$ is a commutative Banach algebra. $\square$

Theorem 1.4.4. *Let $f, g \in \mathcal{C}(\mathbb{T})$. Then $f * g \in \mathcal{C}(\mathbb{T})$.*

Proof. Let $f, g \in \mathcal{C}(\mathbb{T})$. Let $\epsilon > 0$ be given. Let $M = \max_{y \in \mathbb{T}} |g(y)|$. For $s, t \in \mathbb{T}$, we have

$$\begin{aligned} |(f * g)(s) - (f * g)(t)| &= \frac{1}{2\pi} \left| \int_{-\pi}^{\pi} f(s-y) g(y) dy - \int_{-\pi}^{\pi} f(t-y) g(y) dy \right| \\ &\leq \frac{1}{2\pi} \int_{-\pi}^{\pi} |f(s-y) - f(t-y)| |g(y)| dy \\ &\leq \frac{M}{2\pi} \int_{-\pi}^{\pi} |f(s-y) - f(t-y)| dy. \end{aligned}$$

Since f is uniformly continuous on $\mathbb{T}$, there exists $\delta > 0$ such that $|s - t| < \delta$ implies $|f(s) - f(t)| < \frac{\epsilon}{M}$. Thus $|(f * g)(s) - (f * g)(t)| \leq \epsilon$ if $|s - t| < \delta$, proving that $f * g \in \mathcal{C}(\mathbb{T})$. $\square$

In general if $f, g \in L^1(\mathbb{T})$, then we can show that there exist sequences of functions $\{f_n\}, \{g_n\}$ in $\mathcal{C}(\mathbb{T})$ such that $f_n * g_n$ converges to $f * g$ in L^1 . However, if we assume that both f and g are bounded, then the convergence becomes uniform, which is established in the next theorem.

Theorem 1.4.5. *Let $f, g \in L^1(\mathbb{T})$. If, in addition, f and g are bounded, then there exist sequences $\{f_n\}$ and $\{g_n\}$ in $\mathcal{C}(\mathbb{T})$ such that $f_n * g_n$ converges to $f * g$ uniformly on $\mathbb{T}$.*

Proof. Let $f, g \in L^1(\mathbb{T})$ be bounded. Then there exist $\{f_n\}$ and $\{g_n\}$ in $\mathcal{C}(\mathbb{T})$ such that

$$\|f_n - f\|_1 \rightarrow 0 \quad \text{as} \quad n \rightarrow \infty \quad \sup_{x \in \mathbb{T}} |f_n(x)| \leq k_1, \forall n$$

and

$$\|g_n - g\|_1 \rightarrow 0 \quad \text{as } n \rightarrow \infty \sup_{x \in \mathbb{T}} |g_n(x)| \leq k_2, \forall n.$$

Consider

$$\begin{aligned} f * g - f_n * g_n &= f * g - f_n * g + f_n * g - f_n * g_n \\ &= (f - f_n) * g + f_n * (g - g_n). \end{aligned}$$

Let $M = \sup_{y \in \mathbb{T}} |g(y)|$. For $x \in \mathbb{T}$, we have,

$$\begin{aligned} (f - f_n) * g(x) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} (f - f_n)(x - y) g(y) dy \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} (f(x - y) - f_n(x - y)) g(y) dy. \end{aligned}$$

Thus

$$\begin{aligned} |(f - f_n) * g(x)| &\leq \frac{1}{2\pi} \int_{-\pi}^{\pi} |f(x - y) - f_n(x - y)| |g(y)| dy \\ &\leq \frac{1}{2\pi} \sup_{y \in \mathbb{T}} |g(y)| \int_{-\pi}^{\pi} |f(x - y) - f_n(x - y)| dy \\ &\leq \frac{M}{2\pi} \|f_n - f\|_1 \rightarrow 0 \text{ as } n \rightarrow \infty. \end{aligned}$$

Also

$$\begin{aligned} |f_n * (g - g_n)(x)| &= \frac{1}{2\pi} \left| \int_{-\pi}^{\pi} (g - g_n)(x - y) f_n(y) dy \right| \\ &\leq \frac{1}{2\pi} \int_{-\pi}^{\pi} |g(x - y) - g_n(x - y)| |f_n(y)| dy \\ &\leq \frac{k_1}{2\pi} \int_{-\pi}^{\pi} |g(x - y) - g_n(x - y)| dy \end{aligned}$$

$$= \frac{k_1}{2\pi} \|g_n - g\|_1 \rightarrow 0 \text{ as } n \rightarrow \infty.$$

Thus $f_n * g_n - f * g \rightarrow 0$ uniformly on $\mathbb{T}$ as $n \rightarrow \infty$. In other words, $f_n * g_n \rightarrow f * g$ uniformly on $\mathbb{T}$ as $n \rightarrow \infty$. $\square$

Theorem 1.4.6. *Let $f, g \in L^1(\mathbb{T})$. Then $(f * g)\widehat{(\cdot)}(n) = \widehat{f}(n)\widehat{g}(n), \forall n \in \mathbb{Z}$.*

Proof.

$$\begin{aligned} (f * g)\widehat{(\cdot)}(n) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} (f * g)(x) e^{-inx} dx \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} f(x-y) g(y) dy \right) e^{-inx} dx \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} f(x-y) e^{-inx} dx \right) g(y) dy \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \left(\frac{1}{2\pi} e^{-iny} \int_{-\pi}^{\pi} f(x-y) e^{-in(x-y)} dx \right) g(y) dy \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \widehat{f}(n) g(y) e^{-iny} dy = \widehat{f}(n)\widehat{g}(n). \end{aligned} \quad \square$$

1.5 An approximate identity

In the previous section, we have shown that $L^1(\mathbb{T})$ is Banach algebra under convolution. It is therefore natural to ask if $L^1(\mathbb{T})$ admits an identity. In other words, the question is whether there exists $\varphi \in L^1(\mathbb{T})$ such that $f * \varphi = f$ for all $f \in L^1$. It is easy to see that this is not the case. Indeed, if such a φ exists, then by Theorem

1.4.6 $\widehat{f}(n)\widehat{\varphi}(n) = \widehat{f\varphi}(n)$ for all n which is possible only if $\widehat{\varphi}(n) = 1$ for all n . But for $\varphi \in L^1(\mathbb{T})$, $\widehat{\varphi}(n) \rightarrow 0$ as $n \rightarrow \pm\infty$. (See Corollary 1.7.2.) Hence there is no $\varphi \in L^1(\mathbb{T})$ such that $f * \varphi = f$ for all f .

Thus we look for a sequence $\{e_n\}$ such that $f * e_n \rightarrow f$ as $n \rightarrow \infty$ in appropriate norm. Such a sequence is called an approximate identity.

Towards this end, first we shall prove the following theorem.

Theorem 1.5.1. *Let $\{k_n\}$ be a sequence of functions defined on T satisfying*

$$1. \quad \frac{1}{2\pi} \int_{-\pi}^{\pi} k_n(x) dx = 1 \quad \forall n.$$

$$2. \quad \text{There exists } 0 \leq M < \infty \text{ such that } \int_{-\pi}^{\pi} |k_n(x)| dx \leq M, \forall n \geq 1.$$

$$3. \quad \text{For every } \delta > 0, \quad \int_{\delta \leq |x| \leq \pi} |k_n(x)| dx \rightarrow 0 \text{ as } n \rightarrow \infty.$$

If $f \in \mathcal{C}(\mathbb{T})$, then $k_n * f$ converges to f uniformly on T .

Proof. As $\frac{1}{2\pi} \int_{-\pi}^{\pi} k_n(y) dy = 1$,

$$f * k_n(x) - f(x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} [f(x-y) - f(x)] k_n(y) dy,$$

which gives

$$|f * k_n(x) - f(x)| \leq \frac{1}{2\pi} \int_{-\pi}^{\pi} |f(x-y) - f(x)| |k_n(y)| dy. \quad (1.10)$$

Let $\epsilon > 0$ be given. Then by uniform continuity of f there exists $\delta > 0$ such that $|f(s) - f(t)| < \epsilon$ whenever $|s - t| < \delta$. By (1.10),

$$\begin{aligned} |f * k_n(x) - f(x)| &\leq \frac{1}{2\pi} \int_{-\delta}^{\delta} |f(x-y) - f(x)| |k_n(y)| dy \\ &\quad + \frac{1}{2\pi} \int_{\delta \leq |y| \leq \pi} |f(x-y) - f(x)| |k_n(y)| dy \\ &=: I_1 + I_2. \end{aligned}$$

We have,

$$\begin{aligned} I_1 &= \frac{1}{2\pi} \int_{-\delta}^{\delta} |f(x-y) - f(x)| |k_n(y)| dy \\ &\leq \frac{\epsilon}{2\pi} \int_{-\delta}^{\delta} |k_n(y)| dy \leq \frac{\epsilon M}{2\pi}. \end{aligned}$$

On the other hand

$$\begin{aligned} I_2 &= \frac{1}{2\pi} \int_{\delta \leq |y| \leq \pi} |f(x-y) - f(y)| |k_n(y)| dy \\ &\leq \frac{\|f\|_{\infty}}{\pi} \int_{\delta \leq |y| \leq \pi} |k_n(y)| dy, \end{aligned} \tag{1.11}$$

which goes to zero as $n \rightarrow \infty$. Here $\|f\|_{\infty} = \sup_{x \in \mathbb{T}} |f(x)|$. Thus $f * k_n$ converges to f uniformly on $\mathbb{T}$. $\square$

Remark 1.5.2. Since $\mathcal{C}(\mathbb{T})$ is dense in $L^1(\mathbb{T})$ and uniform convergence is stronger than L^1 convergence, it follows from the above theorem that $\{k_n\}$ gives an approximate identity for $L^1(\mathbb{T})$.

Remark 1.5.3. We shall show that the Dirichlet kernel does not satisfy the hypothesis of Theorem 1.5.1. Recall

$$D_N(x) = \sum_{n=-N}^N e^{inx} = \frac{\sin(N + \frac{1}{2})x}{\sin(\frac{x}{2})}.$$

We have $\frac{1}{2\pi} \int_{-\pi}^{\pi} D_N(x) dx = 1$. Since $\frac{1}{2\pi} \int_{-\pi}^{\pi} e^{inx} dx = 0$ for all $n \neq 0$. However, property (ii) of Theorem 1.5.1 fails. We shall show that $\int_{-\pi}^{\pi} |D_N(x)| dx \geq c \ln N, \forall N$.

Consider

$$\begin{aligned} \int_{-\pi}^{\pi} |D_N(x)| dx &= \int_{-\pi}^{\pi} \left| \frac{\sin(N + \frac{1}{2})x}{\sin(\frac{x}{2})} \right| dx \\ &\geq 2 \int_{-\pi}^{\pi} \frac{|\sin(N + \frac{1}{2})x|}{|x|} dx, \end{aligned}$$

since $\left| \frac{\sin \frac{x}{2}}{\frac{x}{2}} \right| \leq 1$. Setting $y = (N + \frac{1}{2})x$, we get,

$$\begin{aligned} \int_{-\pi}^{\pi} |D_N(x)| dx &\geq 2 \int_{-(N+\frac{1}{2})\pi}^{(N+\frac{1}{2})\pi} \left| \frac{\sin y}{y} \right| dy \\ &= 4 \int_0^{(N+\frac{1}{2})\pi} \left| \frac{\sin y}{y} \right| dy \\ &= 4 \left\{ \int_0^{N\pi} \left| \frac{\sin y}{y} \right| dy + \int_{N\pi}^{(N+\frac{1}{2})\pi} \left| \frac{\sin y}{y} \right| dy \right\}. \end{aligned}$$

Since the second term on the right hand side is non-negative, we get,

$$\begin{aligned} \int_{-\pi}^{\pi} |D_N(x)| dx &\geq 4 \int_0^{N\pi} \left| \frac{\sin y}{y} \right| dy \\ &= 4 \sum_{k=0}^{N-1} \int_{k\pi}^{(k+1)\pi} \left| \frac{\sin y}{y} \right| dy. \end{aligned}$$

For $k\pi \leq y \leq (k+1)\pi$, we have $\frac{1}{|y|} \geq \frac{1}{(k+1)\pi}$. Hence

$$\begin{aligned} \int_{-\pi}^{\pi} |D_N(x)| dx &\geq 4 \sum_{k=0}^{N-1} \frac{1}{(k+1)\pi} \int_{k\pi}^{(k+1)\pi} |\sin y| dy \\ &= \frac{4}{\pi} \sum_{k=0}^{N-1} \frac{1}{(k+1)} \int_0^{\pi} |\sin(y)| dy, \\ &\geq c' \sum_{k=0}^{N-1} \frac{1}{(k+1)} \\ &\geq c' \sum_{l=1}^N \frac{1}{l} \geq c'' \log N, \text{ as } N \rightarrow \infty. \end{aligned}$$

1.5.1 Fejér kernel

As an example of approximate identity, we introduce the kernel $\sigma_n(x)$. This is defined, using the Dirichlet kernel D_k , by

$$\sigma_N(x) = \frac{1}{N} \sum_{k=0}^{N-1} D_k(x).$$

Note that

$$\begin{aligned} f * \sigma_N(x) &= \frac{1}{N} \sum_{k=0}^{N-1} f * D_k(x) \\ &= \frac{1}{N} \sum_{k=0}^{N-1} S_n(f(x)), \end{aligned} \tag{1.12}$$

and these are called the Fejér means. Fejér proved the following result. In fact, he has proved for $f \in L^p(\mathbb{T})$, $1 \leq p < \infty$. However, we shall give the proof for $p = 1$ now and give the remaining separately in the solved problem.

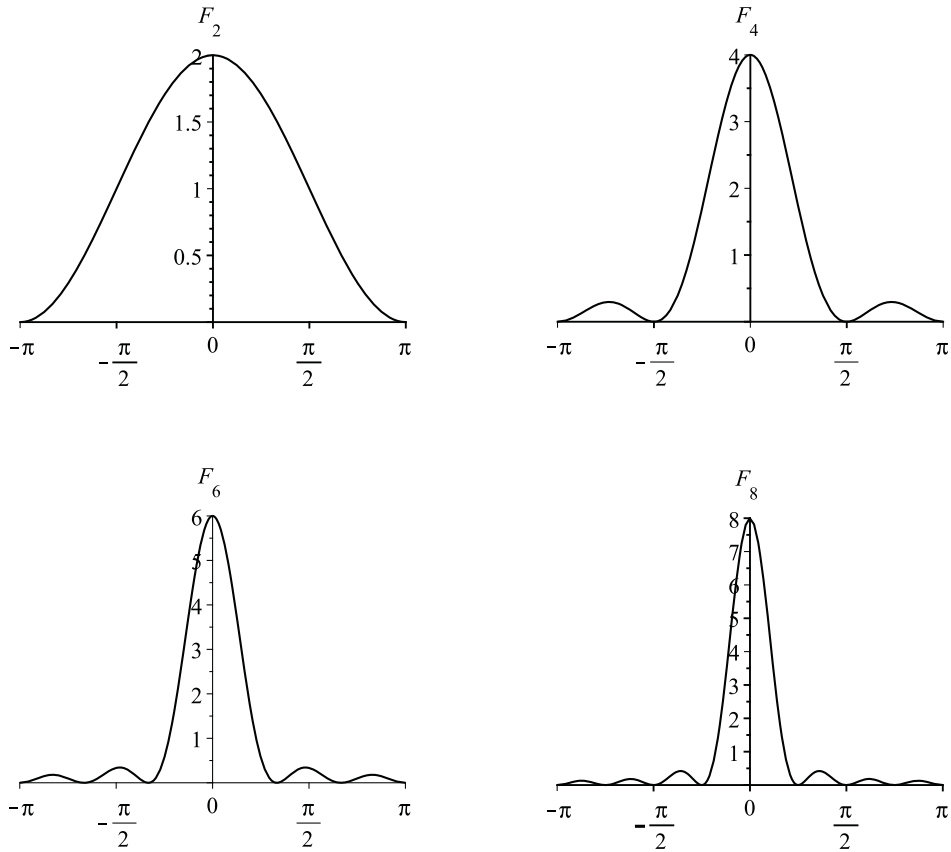


Figure 1.4: Fejér kernel.

Theorem 1.5.4. *Let $f \in L^1(\mathbb{T})$. Then $f * \sigma_N \rightarrow f$ in $L^1(\mathbb{T})$.*

Proof. The theorem will follow if we can show that $\{\sigma_N\}$ is an approximate identity.

In order to show this result, first we shall calculate $N\sigma_N(x)$.

Consider

$$\begin{aligned}
N\sigma_N(x) &= \sum_{n=0}^{N-1} D_n(x) = \sum_{n=0}^{N-1} \left(\frac{w^{-n} - w^{n+1}}{1 - w} \right) \\
&= \frac{1}{1 - w} \left\{ \sum_{n=0}^{N-1} w^{-n} - \sum_{n=0}^{N-1} w^{n+1} \right\} \\
&= \frac{1}{1 - w} \left\{ \frac{1 - w^{-N}}{1 - \frac{1}{w}} - w \frac{1 - w^N}{1 - w} \right\} \\
&= \frac{1}{1 - w} \left\{ w \frac{(1 - w^{-N})}{w - 1} - w \frac{(1 - w^N)}{1 - w} \right\} \\
&= \frac{w}{(1 - w)^2} \{w^{-N} - 1 - 1 + w^N\} \\
&= \frac{w}{(1 - w)^2} (w^N + w^{-N} - 2) \\
&= \frac{w}{(1 - w)^2} (w^{N/2} - w^{-N/2})^2.
\end{aligned}$$

Thus

$$\sigma_N(x) = \frac{1}{N} \left(\frac{\sin\left(N\frac{x}{2}\right)}{\sin\left(\frac{x}{2}\right)} \right)^2.$$

Note that $\sigma_N(x) \geq 0$ for every x . Further

$$\begin{aligned}
\frac{1}{2\pi} \int_{-\pi}^{\pi} \sigma_N(x) dx &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{1}{N} \sum_{n=0}^{N-1} D_n(x) dx \\
&= \frac{1}{2\pi N} \sum_{n=0}^{N-1} \int_{-\pi}^{\pi} D_n(x) dx = 1.
\end{aligned}$$

Fix $\delta > 0$. Let $\delta \leq |x| \leq \pi$. Since $\sin^2\left(\frac{x}{2}\right) \geq \sin^2\left(\frac{\delta}{2}\right) = C_\delta > 0$ we get $\sigma_N(x) \leq \frac{\sin^2\left(N\frac{x}{2}\right)}{N C_\delta}$.

Consider

$$\begin{aligned} \int_{\delta \leq |x| \leq \pi} |\sigma_N(x)| dx &= \frac{1}{N} \int_{\delta \leq |x| \leq \pi} \frac{\sin^2\left(N\frac{x}{2}\right)}{\sin^2\left(\frac{x}{2}\right)} dx \\ &\leq \frac{1}{N} \frac{1}{C_\delta} \int_{\delta \leq |x| \leq \pi} \sin^2\left(N\frac{x}{2}\right) dx. \\ &\leq \frac{1}{N} \frac{2\pi}{C_\delta} \rightarrow 0 \text{ as } N \rightarrow \infty. \end{aligned}$$

Thus σ_N satisfies the required properties of Theorem 1.5.1. Consequently, we conclude that whenever $f \in \mathcal{C}(\mathbb{T})$, then $f * \sigma_N(x)$ converges uniformly to f on T . $\square$

1.5.2 Poisson kernel

Recall that for $0 \leq r < 1$,

$$P_r(\theta) = \frac{1 - r^2}{1 - 2r \cos \theta + r^2} = \sum_{n=-\infty}^{\infty} r^{|n|} e^{in\theta}.$$

We shall show that $\{P_r\}$ gives an approximate identity. First, notice that $P_r(\theta) \geq 0$.

(i)

$$\begin{aligned} \frac{1}{2\pi} \int_{-\pi}^{\pi} P_r(\theta) d\theta &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \left(\sum_{n=-\infty}^{\infty} r^{|n|} e^{in\theta} \right) d\theta \\ &= \frac{1}{2\pi} \sum_{n=-\infty}^{\infty} \int_{-\pi}^{\pi} r^{|n|} e^{in\theta} d\theta \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{2\pi} \sum_{n=-\infty}^{\infty} r^{|n|} \int_{-\pi}^{\pi} e^{in\theta} d\theta \\
&= 1.
\end{aligned}$$

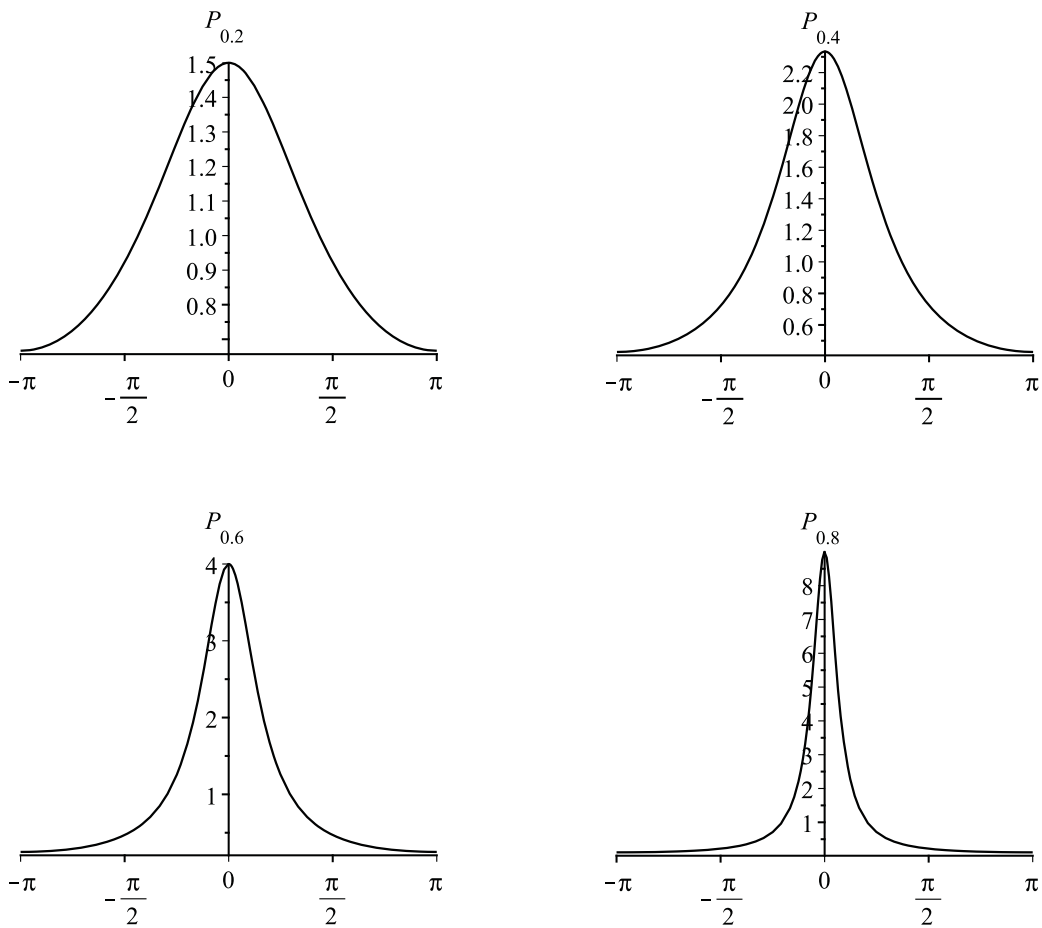


Figure 1.5: Poisson kernel.

(ii) Fix $\delta > 0$ and consider $\delta \leq |\theta| \leq \pi$ and $\frac{1}{2} \leq r < 1$. Then

$$\begin{aligned} 1 - 2r \cos \theta + r^2 &= (1 - r)^2 + 2r(1 - \cos \theta) \\ &= (1 - r)^2 + 4r \sin^2 \left(\frac{\theta}{2} \right) \\ &\geq 2 \sin^2 \left(\frac{\delta}{2} \right) = 2c_\delta. \end{aligned}$$

Hence,

$$\begin{aligned} P_r(\theta) &= \frac{1 - r^2}{1 - 2r \cos \theta + r^2}, \\ &\leq \frac{1 - r^2}{2c_\delta}, \end{aligned}$$

and consequently

$$\int_{\delta \leq |\theta| \leq \pi} P_r(\theta) d\theta \leq \frac{1 - r^2}{2c_\delta} \cdot 2\pi \rightarrow 0 \text{ as } r \rightarrow 1.$$

Thus P_r satisfies the required properties for an approximate identity. Consequently, we conclude that whenever $f \in \mathcal{C}(\mathbb{T})$, $f * P_r(\theta)$ converges uniformly to f on T .

1.6 Summability of Fourier series

Suppose we have a series of complex numbers $\sum_{k=0}^{\infty} c_k$. The n^{th} partial sum of the series is defined as $S_n := \sum_{k=0}^n c_k$. If $S_n \rightarrow s$ as $n \rightarrow \infty$, then we say that $\sum_{k=0}^{\infty} c_k$ converges to s . Let $\sigma_N = \frac{1}{N}(S_0 + S_1 + \cdots + S_{N-1})$. The arithmetic mean σ_N is called the N^{th} partial Cesaro sum of the series $\sum_{k=0}^{\infty} c_k$. If σ_N converges to a limit σ as $N \rightarrow \infty$, then we say that $\sum_{k=0}^{\infty} c_k$ is Cesaro summable to σ .

A series of complex numbers $\sum_{k=0}^{\infty} c_k$ is said to be Abel summable to s if for every $0 \leq r < 1$, the series $\sum_{k=0}^{\infty} c_k r^k$ converges and $\lim_{r \rightarrow 1} \sum_{k=0}^{\infty} c_k r^k = s$.

Remark 1.6.1. Convergence $\Rightarrow$ Cesaro summability $\Rightarrow$ Abel summability. (See more details in Solved problems.)

Consider

$$\begin{aligned}
f * P_r(\theta) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) \left(\sum_{n=-\infty}^{\infty} r^{|n|} e^{in(\theta-x)} \right) dx \\
&= \frac{1}{2\pi} \int_{-\pi}^{\pi} \sum_{n=-\infty}^{\infty} f(x) r^{|n|} e^{in(\theta-x)} dx \\
&= \frac{1}{2\pi} \sum_{n=-\infty}^{\infty} \int_{-\pi}^{\pi} (f(x) e^{-inx}) r^{|n|} e^{in\theta} dx \\
f * P_r(\theta) &= \frac{1}{2\pi} \sum_{n=-\infty}^{\infty} \widehat{f}(n) r^{|n|} e^{in\theta}.
\end{aligned} \tag{1.13}$$

Thus we see that $f * P_r$ are nothing but the Abel means of f . Similarly, $f * \sigma_N$ are nothing but the Cesaro means. Hence we have the following theorems.

Theorem 1.6.2. *If $f \in \mathcal{C}(\mathbb{T})$, then the Fourier series of f is uniformly Cesaro summable to f .*

Theorem 1.6.3. *If $f \in \mathcal{C}(\mathbb{T})$, then the Fourier series of f is uniformly Abel summable to f .*

1.7 Fourier series of $f \in L^2(\mathbb{T})$

Recall that $L^2(\mathbb{T})$ is a Hilbert space with the inner product

$$\langle f, g \rangle = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) \overline{g(t)} dt \quad \text{for } f, g \in L^2(\mathbb{T}).$$

Theorem 1.7.1. (Bessel's inequality) *If $f \in L^2(\mathbb{T})$, then*

$$\sum_{n=-\infty}^{\infty} |\widehat{f}(n)|^2 \leq \frac{1}{2\pi} \int_{-\pi}^{\pi} |f(t)|^2 dt.$$

Proof. Consider

$$\langle f - S_N f, f - S_N f \rangle = \langle f, f \rangle - \langle f, S_N f \rangle - \langle S_N f, f \rangle + \langle S_N f, S_N f \rangle \quad (1.14)$$

Recall that $S_N f(x) = \sum_{n=-N}^N \widehat{f}(n) e^{inx}$. If we denote $e_n(x) = e^{inx}$, $x \in \mathbb{R}$, then we can write $S_N f = \sum_{n=-N}^N \widehat{f}(n) e_n$ and also $\widehat{f}(n) = \langle f, e_n \rangle$. Thus

$$\begin{aligned} \langle f, S_N f \rangle &= \left\langle f, \sum_{n=-N}^N \widehat{f}(n) e_n \right\rangle \\ &= \sum_{n=-N}^N \overline{\widehat{f}(n)} \langle f, e_n \rangle \\ &= \sum_{n=-N}^N \overline{\widehat{f}(n)} \widehat{f}(n) \\ &= \sum_{n=-N}^N |\widehat{f}(n)|^2. \end{aligned}$$

Further

$$\begin{aligned} \langle S_N f, S_N f \rangle &= \left\langle \sum_{n=-N}^N \widehat{f}(n) e_n, \sum_{m=-N}^N \widehat{f}(m) e_m \right\rangle \\ &= \sum_{n=-N}^N \sum_{m=-N}^N \widehat{f}(n) \overline{\widehat{f}(m)} \langle e_n, e_m \rangle. \end{aligned}$$

As $\langle e_n, e_m \rangle = \delta_{n,m}$, we can conclude that

$$\langle S_N f, S_N f \rangle = \sum_{n=-N}^N |\widehat{f}(n)|^2.$$

Hence, (1.14) turns out to be

$$\langle f - S_N f, f - S_N f \rangle = \langle f, f \rangle - \sum_{n=-N}^N |\widehat{f}(n)|^2 \geq 0,$$

from which it follows that

$$\begin{aligned} \sum_{n=-N}^N |\widehat{f}(n)|^2 &\leq \langle f, f \rangle \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} |f(t)|^2 dt. \end{aligned}$$

Letting $N \rightarrow \infty$, we get

$$\sum_{n=-\infty}^{\infty} |\widehat{f}(n)|^2 \leq \frac{1}{2\pi} \int_{-\pi}^{\pi} |f(t)|^2 dt.$$

This leads us to the following important result. □

Corollary 1.7.2. (Riemann-Lebesgue lemma) *If $f \in L^1(\mathbb{T})$, then $\widehat{f}(n) \rightarrow 0$ as $n \rightarrow \pm \infty$. Equivalently, if $f \in L^1(\mathbb{T})$, then $\int_{-\pi}^{\pi} f(x) \cos nx \, dx \rightarrow 0$ and $\int_{-\pi}^{\pi} f(x) \sin nx \, dx \rightarrow 0$ as $n \rightarrow \pm \infty$.*

Proof. The result follows immediately from the theorem for $f \in L^2(\mathbb{T})$. Let $f \in L^1(\mathbb{T})$. Since $L^2(\mathbb{T}) = L^1(\mathbb{T}) \cap L^2(\mathbb{T})$ is dense in $L^1(\mathbb{T})$, there exists a sequence of functions $\{f_n\}$ in $L^2(\mathbb{T})$ such that $f_n \rightarrow f$ in L^1 -norm. Now $\lim_{m \rightarrow \infty} \widehat{f}_n(m) = 0$ for each n . Consider

$$|\widehat{f}_n(m) - \widehat{f}(m)| \leq \frac{1}{2\pi} \int_0^{2\pi} |f_n(x) - f(x)| \, dx$$

$$= \|f_n - f\|_1 \rightarrow 0,$$

as $n \rightarrow \infty$. This shows that $\widehat{f}_n$ converges to $\widehat{f}$ uniformly on $\mathbb{Z}$.

Hence

$$\begin{aligned} \lim_{m \rightarrow \infty} \widehat{f}(m) &= \lim_{m \rightarrow \infty} \lim_{n \rightarrow \infty} \widehat{f}_n(m) \\ &= \lim_{n \rightarrow \infty} \lim_{m \rightarrow \infty} \widehat{f}_n(m) = 0. \end{aligned} \quad \square$$

Theorem 1.7.3. (Best approximation theorem) *Let $f \in L^2(\mathbb{T})$. Let $S_N(f)$ denote the N^{th} partial sum of the Fourier series, and $t_N(x) = \sum_{n=-N}^N c_n e^{inx}$ be any trigonometric polynomial. Then $\|f - S_N f\|_2 \leq \|f - t_N\|_2$ and the equality holds iff $\widehat{f}(n) = c_n$.*

Proof. Consider

$$\|f - t_N\|_2^2 = \langle f - t_N, f - t_N \rangle = \langle f, f \rangle - \langle f, t_N \rangle - \langle t_N, f \rangle + \langle t_N, t_N \rangle. \quad (1.15)$$

But

$$\begin{aligned} \langle f, t_N \rangle &= \left\langle f, \sum_{n=-N}^N c_n e_n \right\rangle \\ &= \sum_{n=-N}^N \bar{c}_n \langle f, e_n \rangle \\ &= \sum_{n=-N}^N \bar{c}_n \widehat{f}(n). \end{aligned}$$

Also

$$\langle t_N, t_N \rangle = \left\langle \sum_{n=-N}^N c_n e_n, \sum_{m=-N}^N c_m e_m \right\rangle$$

$$\begin{aligned}
&= \sum_{n=-N}^N \sum_{m=-m}^m c_n \bar{c}_m \langle e_n, e_m \rangle \\
&= \sum_{n=-N}^N \sum_{m=-m}^m c_n \bar{c}_m \delta_{n,m} \\
&= \sum_{n=-N}^N |c_n|^2.
\end{aligned}$$

Hence (1.15) becomes

$$\begin{aligned}
\|f - t_N\|_2^2 &= \langle f, f \rangle - \sum_{n=-N}^N \bar{c}_n \widehat{f}(n) - \sum_{n=-N}^N c_n \overline{\widehat{f}(n)} + \sum_{n=-N}^N |c_n|^2 \\
&= \langle f, f \rangle - \sum_{n=-N}^N |\widehat{f}(n)|^2 + \sum_{n=-N}^N |\widehat{f}(n) - c_n|^2.
\end{aligned}$$

Thus

$$\|f - t_N\|_2^2 = \|f - S_N f\|_2^2 + \sum_{n=-N}^N |\widehat{f}(n) - c_n|^2,$$

proving our assertion. Recall that a complex valued function f on $\mathbb{T}$ is said to be Lipschitz continuous on $\mathbb{T}$ if $|f(s) - f(t)| < |s - t|$ for all $s, t \in \mathbb{T}$. In the following theorem, we show that Lipschitz continuity leads to the uniform convergence of Fourier series. $\square$

Theorem 1.7.4. *If f is a Lipschitz continuous function on $\mathbb{T}$, then $S_N(f)$ converges to f uniformly on $\mathbb{T}$.*

Proof. Consider $\sin(N + \frac{1}{2})t = \frac{1}{2i} \left(e^{iNt} e^{\frac{it}{2}} - e^{-iNt} e^{\frac{-it}{2}} \right)$. As f is Lipschitz continuous, the function $\frac{f(x-t) - f(x)}{\sin \frac{t}{2}}$ is bounded. Now, applying Riemann-Lebesgue lemma to $\frac{f(x-t) - f(x)}{\sin \frac{t}{2}} e^{\frac{i}{2}t}$ and $\frac{f(x-t) - f(x)}{\sin \frac{t}{2}} e^{\frac{-i}{2}t}$, we can conclude the result. $\square$

Theorem 1.7.5. *Let $f \in \mathcal{C}(\mathbb{T})$. Suppose f is piecewise continuously differentiable function on $\mathbb{T}$. Then the Fourier series of f converges uniformly to f on $\mathbb{T}$.*

Proof. It is enough to prove that $\widehat{f} \in l^1(\mathbb{Z})$. Given that f is a piecewise continuously differentiable function on $\mathbb{T}$. We know that $\mathbb{T}$ can be identified with $[-\pi, \pi]$. Then there exists a partition $x_1, x_2, \dots, x_{m-1} \in [-\pi, \pi]$, such that $-\pi = x_0 \leq x_1 \leq \dots \leq x_{m-1} \leq x_m = \pi$ and $f|_{[x_{j-1}, x_j]}$ is continuously differentiable for each $j = 1, 2, \dots, m$.

Let g_j denote the continuous derivative of f on $[x_{j-1}, x_j]$. Choose a periodic function g on $[-\pi, \pi]$ such that $g = g_j$ on $[x_{j-1}, x_j]$. Then, clearly, $g \in L^2(\mathbb{T})$. Using Bessel's inequality, we have $\sum_{n=-\infty}^{\infty} |\widehat{g}(n)|^2 \leq \|g\|_2^2$.

Consider, for $n \neq 0$

$$\int_{x_{j-1}}^{x_j} f(x) e^{-inx} dx = f(x) \frac{e^{-inx}}{-in} \Big|_{x_{j-1}}^{x_j} + \frac{1}{in} \int_{x_{j-1}}^{x_j} e^{-inx} f'(x) dx.$$

Thus

$$\begin{aligned} \widehat{f}(n) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx \\ &= \frac{1}{2\pi} \sum_{j=1}^m \int_{x_{j-1}}^{x_j} f(x) e^{-inx} dx. \end{aligned}$$

Now

$$\begin{aligned} |\widehat{f}(n)| &= \left| \frac{\widehat{g}(n)}{in} \right| \\ &\leq \frac{1}{2} \left[\frac{1}{n^2} + |\widehat{g}(n)|^2 \right]. \end{aligned}$$

Thus, we get,

$$\sum_{n=-\infty}^{\infty} |\widehat{f}(n)| = |\widehat{f}(0)| + \sum_{n \neq 0} |\widehat{f}(n)|$$

$$\begin{aligned} &\leq |\widehat{f}(0)| + \frac{1}{2} \sum_{n \neq 0} \left(\frac{1}{n^2} + |\widehat{g}(n)|^2 \right) \\ &< \infty, \end{aligned}$$

from which it follows that $\widehat{f} \in l^1(Z)$. □

Theorem 1.7.6. *For any $f \in L^2(\mathbb{T})$, $S_N(f)$ converges to f in $L^2(\mathbb{T})$, i.e. $\|S_N f - f\|_2 \rightarrow 0$ as $N \rightarrow \infty$.*

Proof. Let $\epsilon > 0$ be given. We know that $\mathcal{C}(\mathbb{T})$ is dense in $L^2(\mathbb{T})$. Therefore, given $f \in L^2(\mathbb{T})$, there exists $h \in \mathcal{C}(\mathbb{T})$ such that

$$\|f - h\|_2 < \epsilon. \tag{1.16}$$

Writing $f - S_N f = (f - h) + (h - S_N h) + S_N(h - f)$ and noting that $\|S_N(h - f)\|_2 \leq \|f - h\|_2$, it is enough to show that $\|h - S_N h\|_2 < \epsilon$ for all $N \geq N_0$ for some N_0 . As $h \in \mathcal{C}(\mathbb{T})$, there exists a trigonometric polynomial P of degree N_0 such that $\|h - P\|_2 < \epsilon$. For $N \geq N_0$, $S_N P = P$ and hence

$$\|h - S_N h\|_2 \leq \|h - P\|_2 + \|S_N(P - h)\|_2 \leq 2\epsilon.$$

This proves our assertion. □

Theorem 1.7.7. (Parseval's Theorem) *For any $f, g \in L^2(\mathbb{T})$,*

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) \overline{g(x)} dx = \sum_{n=-\infty}^{\infty} \widehat{f}(n) \overline{\widehat{g}(n)}.$$

In particular,

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} |f(x)|^2 dx = \sum_{n=-\infty}^{\infty} |\widehat{f}(n)|^2.$$

Proof. We have shown in Theorem 1.7.6 that $S_N f \rightarrow f$ in L^2 . Hence, $\langle S_N f, S_N g \rangle \rightarrow \langle f, g \rangle$ which simply means

$$\sum_{n=-\infty}^{\infty} \widehat{f}(n) \overline{\widehat{g}(n)} = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) \overline{g(x)} dx. \quad \square$$

Corollary 1.7.8. *The functions $\{e^{inx} : n \in \mathbb{Z}\}$ forms an orthonormal basis for $L^2(\mathbb{T})$.*

(See Appendix A.1.3.)

Corollary 1.7.9. *Let $f \in L^2(\mathbb{T})$. Then the Fourier series of f can be integrated term by term in $(a, b) \subseteq [-\pi, \pi]$.*

Proof. Let $g(t) = \chi_{[a,b]}(t)$. Then $g \in L^2(\mathbb{T})$ and hence by Parseval's theorem,

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) \overline{g(t)} dt = \sum_{n=-\infty}^{\infty} \widehat{f}(n) \overline{\widehat{g}(n)}.$$

This means that

$$\frac{1}{2\pi} \int_a^b f(t) dt = \frac{1}{2\pi} \sum_{n=-\infty}^{\infty} \widehat{f}(n) \left(\int_a^b e^{int} dt \right),$$

which is our assertion. $\square$

1.8 Existence of a continuous function whose Fourier series diverges

We shall show in this section that, there exists a continuous function whose Fourier series diverges at 0. In order to prove the theorem, we make use of the uniform boundedness principle from functional analysis. (See Appendix A.2.5.)

Let $X = C[-\pi, \pi]$, $Y = \mathbb{R}$. Define $A_N f := S_N f(0)$. Then $|A_N f| = |S_N f(0)| = |D_N * f(0)| \leq \|f\|_\infty \|D_N\|_1$. In other words, the operator norm of A_N is given by $\|A_N\| \leq \|D_N\|_1$. We shall show that $\|A_N\| = \|D_N\|_1$. Take $g(x) = \text{sgn} D_N(x)$. Then $\|g\|_\infty = 1$. Although g is not continuous, it can be approximated by a sequence of continuous functions g_n .

Choose g_n such that $\|g_n\|_\infty = 1$. Then $A_N g_n \rightarrow \|D_N\|_1$, by using Lebesgue dominated convergence theorem. Thus, $\|A_N\| = \|D_N\|_1$. But $\|D_N\|_1 \geq c \log N$ (see Remark 1.5.3). Hence, $\|A_N\| \rightarrow \infty$ as $N \rightarrow \infty$. Hence, it follows from uniform boundedness principle that there exists a continuous function f in $[-\pi, \pi]$ such that $|A_N f| = |S_N f(0)| \rightarrow \infty$ as $N \rightarrow \infty$, which means the Fourier series of f diverges at 0.

1.9 Brief history of Fourier series

The appearance of Fourier series can be traced back to the works of d'Alembert (1747), Euler (1748) and D. Bernoulli (1753) in the study of vibrating strings, but the theory of Fourier series truly began with the profound work of Fourier on heat conduction at the beginning of the nineteenth century. He studied the problem of heat flow in a thin wire fixed at two end points with length π and zero temperature

at the ends. He showed that the initial temperature can be expressed as the sum of an infinite series of sines and cosines, now popularly known as Fourier series. Even though Fourier did not give a convincing proof of convergence of such infinite series, he offered the conjecture that convergence holds for an arbitrary function. Subsequent work by Dirichlet, Riemann, Lebesgue, and others, throughout the next two hundred years was needed to show that any arbitrary periodic function can be expressed in terms of trigonometric series.

1.10 Solved problems

1.10.1. Suppose that $f \in C^k(\mathbb{T})$. Show that $\widehat{f}(n) = O\left(\frac{1}{|n|^k}\right)$ as $n \rightarrow \infty$.

Solution.

$$\begin{aligned}\widehat{f}(n) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) d\left(\frac{e^{-inx}}{-in}\right).\end{aligned}$$

Using Bernoulli's formula for successive integration by parts, we get

$$\widehat{f}(n) = \frac{(-1)^k}{(-in)^k} (f^{(k)})^\wedge(n).$$

Then

$$\begin{aligned}|\widehat{f}(n)| &\leq \frac{1}{|n|^k} |(f^{(k)})^\wedge(n)| \\ &\leq \frac{1}{|n|^k} \|f^{(k)}\|_1.\end{aligned}$$

1.10.2. If $f \in L^1(\mathbb{T})$, then show that $\widehat{f}(n) = \frac{1}{4\pi} \int_{-\pi}^{\pi} (f(x) - f(x + \pi/n)) e^{-inx} dx$. In

addition, if f satisfies the Holder's condition of order α , then $\widehat{f}(n) = O\left(\frac{1}{|n|^\alpha}\right)$.

Solution. We have,

$$\widehat{f}(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx.$$

Since $e^{i\pi} = -1$, we can write

$$\begin{aligned} \widehat{f}(n) &= \frac{-1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx+\pi i} dx \\ &= \frac{-1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-in(x-\frac{\pi}{n})} dx. \\ &= \frac{-1}{2\pi} \int_{-\pi-\pi/n}^{\pi-\pi/n} f(y + \frac{\pi}{n}) e^{-iny} dy. \end{aligned}$$

Hence

$$\widehat{f}(n) = \frac{-1}{2\pi} \int_{-\pi}^{\pi} f(x + \frac{\pi}{n}) e^{-inx} dx.$$

Thus

$$2\widehat{f}(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} - \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x + \frac{\pi}{n}) e^{-inx} dx,$$

from which it follows that,

$$\widehat{f}(n) = \frac{1}{4\pi} \int_{-\pi}^{\pi} (f(x) - f(x + \frac{\pi}{n})) e^{-inx} dx.$$

Then

$$\begin{aligned}
|\widehat{f}(n)| &\leq \frac{1}{4\pi} \int_{-\pi}^{\pi} (f(x) - f(x + \frac{\pi}{n})) dx \\
&\leq \frac{1}{4\pi} \int_{-\pi}^{\pi} c \left| \frac{\pi}{n} \right|^{\alpha} \cdot 2\pi \\
&= \frac{k}{|n|^{\alpha}},
\end{aligned}$$

where k is a constant.

1.10.3. Using Riemann-Lebesgue lemma, prove that $\lim_{R \rightarrow \infty} \int_0^R \frac{\sin x}{x} dx = \frac{\pi}{2}$.

Solution. Let $I = \lim_{R \rightarrow \infty} \int_0^R \frac{\sin x}{x} dx$. Let $R = (N + \frac{1}{2})\pi$, then

$$I = \lim_{N \rightarrow \infty} \int_0^{(N + \frac{1}{2})\pi} \frac{\sin x}{x} dx.$$

Applying change of variable $x = (N + \frac{1}{2})y$, we get

$$I = \lim_{N \rightarrow \infty} \int_0^{\pi} \frac{\sin (N + \frac{1}{2})y}{y} dy.$$

From the property of Dirichlet kernel, we get

$$\int_0^{\pi} \frac{\sin (N + \frac{1}{2})y}{\sin (\frac{y}{2})} dy = \pi$$

Hence,

$$\begin{aligned} I - \pi/2 &= \lim_{N \rightarrow \infty} \int_0^\pi \left(\frac{\sin(N + \frac{1}{2})y}{y} dy - \frac{\sin(N + \frac{1}{2})y}{2 \sin(\frac{y}{2})} \right) dy \\ &= \lim_{N \rightarrow \infty} \int_0^\pi \left(\frac{1}{y} - \frac{1}{2 \sin(\frac{y}{2})} \right) \sin(N + \frac{1}{2})y \, dy. \end{aligned}$$

Since, $\frac{1}{y} - \frac{1}{2 \sin(\frac{y}{2})} \in L^2(\mathbb{T})$, from Riemann-Lebesgue lemma, it follows that $I = \pi/2$.

1.10.4. Prove the Binomial identity $\sum_{k=0}^n \binom{n}{k}^2 = \binom{2n}{n}$.

Solution. Consider

$$\sum_{k=0}^n \binom{n}{k} e_k(x) = (1 + e_1(x))^n.$$

By Parseval's inequality

$$\begin{aligned} \sum_{k=0}^n \binom{n}{k}^2 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} |f(x)|^2 dx = \frac{1}{2\pi} \int_{-\pi}^{\pi} |(1 + e_1(x))^n| dx \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} (1 + e_1(x))^n \overline{(1 + e_1(x))^n} dx \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} (1 + e_1(x))^n (1 + \overline{e_1(x)})^n dx \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} (1 + e_1(x))(1 + \overline{e_1(x)})^n dx \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} (2 + 2 \cos x)^n dx \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{2\pi} \cdot 2^n \int_{-\pi}^{\pi} (1 + \cos x)^n dx \\
&= \frac{2^{2n}}{2\pi} \cdot \int_{-\pi}^{\pi} \cos^{2n} \left(\frac{x}{2} \right) dx \\
&= \binom{2n}{k}.
\end{aligned}$$

1.10.5. By considering the Fourier series of $f(x) = |x|$, $x \in [-\pi, \pi]$, find the sum of the series $\sum_{n=0}^{\infty} \frac{1}{(2n+1)^4}$ and $\sum_{n=0}^{\infty} \frac{1}{n^4}$. What is the sum of $\sum_{n=1}^{\infty} \frac{1}{n^2}$.

Solution. We have

$$\begin{aligned}
\widehat{f}(n) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} |x| e^{-inx} dx \\
&= \frac{1}{2\pi} \left\{ \int_{-\pi}^{\pi} |x| \cos nx dx - i \int_{-\pi}^{\pi} |x| \sin nx dx \right\} \\
&= \frac{1}{2\pi} \left\{ 2 \int_0^{\pi} x \cos nx dx \right\} \\
&= \frac{1}{\pi} \left\{ \left. \frac{x \sin nx}{n} \right|_0^{\pi} - \int_0^{\pi} \frac{\sin nx}{n} dx \right\} \quad \text{if } n \neq 0 \\
&= -\frac{1}{\pi} \cdot \frac{1}{n} \int_0^{\pi} \sin nx dx \\
&= \frac{1}{n^2 \pi} \cos nx \Big|_0^{\pi} \\
&= \frac{1}{n^2 \pi} \{(-1)^n - 1\}.
\end{aligned}$$

For $n \neq 0$, we have

$$\widehat{f}(n) = \begin{cases} 0 & \text{if } n \text{ is even} \\ \frac{-2}{n^2\pi} & \text{if } n \text{ is odd} \end{cases}$$

$$\begin{aligned} \widehat{f}(0) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} |x| dx \\ &= \frac{1}{\pi} \int_0^{\pi} x dx \\ &= \frac{1}{\pi} \left(\frac{x^2}{2} \right)_0^{\pi} \\ &= \frac{\pi}{2}. \end{aligned}$$

Thus the Fourier Series of $f(x) = |x|$ on $[-\pi, \pi]$ is given by

$$\begin{aligned} |x| &= \widehat{f}(0) + \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} \widehat{f}(n) e^{inx} \\ &= \frac{\pi}{2} + \sum_{n=\pm 1, \pm 3, \dots} \frac{-2}{n^2\pi} e^{inx} \\ &= \frac{\pi}{2} + 2(-2) \sum_{n=1}^{\infty} \frac{1}{\pi(2n-1)^2} e^{i(2n-1)x}, \quad x \in [-\pi, \pi] \\ |x| &= \frac{\pi}{2} - \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} e^{i(2n-1)x}, \quad x \in [-\pi, \pi] \end{aligned}$$

On putting $x = 0$, we get

$$0 = \frac{\pi}{2} - \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2}$$

Hence, it follows that

$$\sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} = \frac{\pi^2}{8}.$$

Now we shall calculate the sum of the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$. Let $s = \sum_{n=1}^{\infty} \frac{1}{n^2}$. Then

$$\begin{aligned} s &= \sum_{n=2,4}^{\infty} \frac{1}{n^2} + \sum_{n=1,3}^{\infty} \frac{1}{n^2} \\ &= \sum_{m \in \mathbb{N}} \frac{1}{(2m)^2} + \sum_{m \in \mathbb{N}} \frac{1}{(2m-1)^2} \\ &= \frac{1}{4} \sum_{m=1}^{\infty} \frac{1}{m^2} + \frac{\pi^2}{8} \end{aligned}$$

Thus

$$s = \frac{1}{4}s + \frac{\pi^2}{8}.$$

This leads to

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

Using Parseval's inequality, we have

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} x^2 dx = \frac{\pi^2}{4} + \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} |\hat{f}(n)|^2.$$

Hence,

$$\frac{1}{\pi} \int_0^{\pi} x^2 dx = \frac{\pi^2}{4} + \sum_{n=\pm 1, \pm 3, \dots} \frac{4}{n^4 \pi^2},$$

which means that

$$\frac{1}{\pi} \cdot \frac{\pi^3}{3} = \frac{\pi^2}{4} + \frac{8}{\pi^2} \sum_{n=1,3,\dots} \frac{1}{n^4}.$$

Thus it follows that

$$\frac{\pi^2}{3} - \frac{\pi^2}{4} = \frac{8}{\pi^2} \sum_{n=0,1,\dots} \frac{1}{(2n+1)^4},$$

from which we arrive at

$$\frac{\pi^2}{12} = \frac{8}{\pi^2} \sum_{n=0,1,\dots} \frac{1}{(2n+1)^4}.$$

This implies that

$$\sum_{n=0}^{\infty} \frac{1}{(2n+1)^4} = \frac{\pi^4}{96}.$$

Next, we shall calculate the sum $\sum_{n=1}^{\infty} \frac{1}{n^4}$.

Let

$$s' = \sum_{n=1}^{\infty} \frac{1}{n^4}.$$

Then

$$\begin{aligned} s' &= \sum_{n \text{ even}} \frac{1}{n^4} + \sum_{n \text{ odd}} \frac{1}{n^4} \\ &= \sum_{n=1}^{\infty} \frac{1}{(2n)^4} + \sum_{n=0}^{\infty} \frac{1}{(2n+1)^4} \\ &= \frac{1}{16} \sum_{n=1}^{\infty} \frac{1}{n^4} + \sum_{n=0}^{\infty} \frac{1}{(2n+1)^4}. \end{aligned}$$

Hence

$$s' - \frac{1}{16}s' = \frac{\pi^4}{96}.$$

This leads to

$$\sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}.$$

1.10.6. Compute the Fourier coefficients of the function f given by

$$f(x) = \begin{cases} |x| & |x| < 1 \\ 0 & \text{otherwise.} \end{cases}$$

Solution. We have

$$\begin{aligned} \hat{f}(n) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx \\ &= \frac{1}{2\pi} \int_{-1}^1 |x| e^{-inx} dx \\ &= \frac{1}{2\pi} \left\{ \int_{-1}^0 -x e^{-inx} dx + \int_0^1 x e^{-inx} dx \right\} \\ &= -\frac{1}{2\pi} \left\{ \left. \frac{x e^{-inx}}{-in} \right|_{-1}^0 - \int_{-1}^0 \frac{e^{-inx}}{-in} dx - \left. \frac{x e^{-inx}}{-in} \right|_0^1 + \int_0^1 \frac{e^{-inx}}{-in} dx \right\} \\ &= -\frac{1}{2\pi} \left\{ \frac{e^{in}}{(-in)} + \frac{1}{in} \left(\frac{e^{-inx}}{-in} \right)_{-1}^0 - \frac{e^{-in}}{(-in)} + \frac{-1}{in} \left(\frac{e^{-inx}}{-in} \right)_0^1 \right\} \\ &= -\frac{1}{2\pi} \left\{ \frac{1}{in} (e^{-in} - e^{in}) + \frac{1}{n^2} - \frac{e^{in}}{n^2} - \frac{e^{-in}}{n^2} + \frac{1}{n^2} \right\} \\ &= -\frac{1}{2\pi} \left\{ \frac{1}{in} (e^{-in} - e^{in}) + \frac{2}{n^2} - \frac{1}{n^2} (e^{in} + e^{-in}) \right\} \\ &= -\frac{1}{2\pi} \left\{ \frac{2}{n^2} - \frac{2 \sin n}{n} - \frac{2 \cos n}{n^2} \right\} \end{aligned}$$

$$= -\frac{1}{n\pi} \left\{ \frac{1}{n} - \sin n - \frac{\cos n}{n} \right\}.$$

1.10.7. Compute the Fourier coefficients of 1-periodic function $f(t) = 1$, $0 \leq t \leq 1$.

Solution. We have

$$\begin{aligned} \widehat{f}(n) &= \int_0^1 e^{-2\pi i n t} dt \\ &= \frac{e^{-2\pi i n t}}{-2\pi i n} \Big|_0^1 \\ &= \frac{i}{2\pi n} [e^{-2\pi i n} - 1] \\ &= \frac{i}{2\pi n} [(-1)^n - 1], \quad n \neq 0 \end{aligned}$$

But

$$\widehat{f}(0) = \int_0^1 dt = 1.$$

Hence

$$\widehat{f}(n) = \begin{cases} \frac{i}{2\pi n} [(-1)^n - 1] & n \neq 0 \\ 1 & n = 0. \end{cases}$$

1.10.8. Let $f \in L^1(\mathbb{T})$ such that $\widehat{f}(0) = 0$. Define $F(t) = \int_0^t f(\tau) d\tau$. Show that F is continuous, 2π -periodic and $\widehat{F}(n) = \frac{\widehat{f}(n)}{in}$, $n \neq 0$.

Solution. Let $\epsilon > 0$ be given. Then for $t > s$,

$$|F(t) - F(s)| = \left| \int_0^t f(\tau) d\tau - \int_0^s f(\tau) d\tau \right|$$

$$\begin{aligned}
&= \left| \int_s^t f(\tau) d\tau \right| \\
&\leq \int_s^t |f(\tau)| d\tau.
\end{aligned}$$

As $f \in L^1(\mathbb{T})$, there exists a $\delta > 0$ such that

$$\int_s^t |f(\tau)| d\tau \leq \epsilon \text{ whenever } |t - s| < \delta.$$

Thus, F is continuous. Consider

$$\begin{aligned}
F(t + 2\pi) &= \int_0^{t+2\pi} f(\tau) d\tau \\
&= \int_0^{2\pi} f(\tau) d\tau + \int_{2\pi}^{2\pi+t} f(\tau) d\tau \\
&= 2\pi \widehat{f}(0) + \int_0^t f(\tau) d\tau \\
&= F(t).
\end{aligned}$$

Thus F is 2π -periodic. By Radon-Nikodym theorem, F is differentiable with $F'(t) = f(t) - f(0)$. Thus

$$\begin{aligned}
\widehat{F}(n) &= \int_0^{2\pi} F(t) e^{-int} dt \\
&= \int_0^{2\pi} F(t) d\left(\frac{e^{-int}}{-in}\right)
\end{aligned}$$

$$\begin{aligned}
&= F(t) \frac{e^{-int}}{-in} \Big|_0^{2\pi} - \int_0^{2\pi} \frac{e^{-int}}{-in} F'(t) dt \\
&= \frac{1}{in} \int_0^{2\pi} f(t) e^{-int} dt - f(0) \int_0^{2\pi} e^{-int} dt \\
&= \frac{1}{in} \widehat{f}(n), \quad n \neq 0.
\end{aligned}$$

1.10.9. Calculate $\chi_{[-\pi,0]} * \chi_{[0,\pi]}$ in $L^1(\mathbb{T})$.

Solution. Note that

$$\chi_{[-\pi,0]} * \chi_{[0,\pi]}(t) = \frac{1}{2\pi} \int_{-\pi}^0 \chi_{[0,\pi]}(t-s) ds.$$

Now, for this integral to be non-zero, s should belong to $[-\pi, 0]$ and $[-\pi+t, t]$.

If $-\pi \leq t \leq 0$

$$\chi_{[-\pi,0]} * \chi_{[0,\pi]}(t) = \frac{1}{2\pi} \int_{-\pi}^t dt = \frac{1}{2\pi} [t + \pi].$$

If $0 < t \leq \pi$, then

$$\chi_{[-\pi,0]} * \chi_{[0,\pi]}(t) = \frac{1}{2\pi} \int_{-\pi+t}^0 dt = \frac{1}{2\pi} [-t + \pi].$$

Hence

$$\chi_{[-\pi,0]} * \chi_{[0,\pi]}(t) = \frac{1}{2\pi} (\pi - |t|) \chi_{[-\pi,\pi]}(t).$$

1.10.10. Suppose $\theta : \mathbb{R} \rightarrow \mathbb{T}$ is a continuous group homomorphism then show that $\theta(t) = e^{itx}$ for some real number x . Use this to derive the form of the continuous group homomorphism $\theta : \mathbb{T} \rightarrow \mathbb{T}$.

Solution. If θ is the trivial homomorphism, then we are done. If not, there exists a $a > 0$ such that

$$c = \int_0^a \theta(t) dt \neq 0.$$

Thus

$$c\theta(x) = \int_0^a \theta(x+t) dt = \int_0^{a+x} \theta(t) dt.$$

So, θ is differentiable and

$$\theta'(x) = \frac{1}{c}[\theta(a+x) - \theta(x)] = \frac{1}{c}[\theta(a) - 1]\theta(x) = c',$$

where $c' = \frac{1}{c}[\theta(a) - 1]$. Thus, θ satisfies the ordinary differential equation $\theta' = c'\theta$. Hence, $\theta(t) = e^{c't}$ and since $|\theta(t)| = 1 \forall t$, $c' = 2\pi i x$ for some $x \in \mathbb{R}$.

Since $\mathbb{T} \simeq \mathbb{R}/\mathbb{Z}$, any continuous group homomorphism $\theta : \mathbb{T} \rightarrow \mathbb{T}$ gives rise to a continuous group homomorphism $\theta' : \mathbb{R} \rightarrow \mathbb{T}$ given by $\theta' = \theta \cdot \pi$, where π is the canonical quotient map, $\pi : \mathbb{R} \rightarrow \mathbb{R}/\mathbb{Z}$. Also, $\ker \theta'$ contains $\mathbb{Z}$. Thus by the form of the continuous homomorphism from $\mathbb{R} \rightarrow \mathbb{T}$, $\theta(t) = e^{int}$ for some integer n .

1.10.11. Let G be a closed proper subgroup of $\mathbb{T}$. Show that G is finite.

Solution. Since G is closed, $\mathbb{T}/G$ is an abelian group. Let $\theta : \mathbb{T}/G \rightarrow \mathbb{T}$ be a continuous group homomorphism. Then the map $\theta' : \mathbb{T} \rightarrow \mathbb{T}$ defined as $\theta' = \theta \cdot \pi$, $\pi : \mathbb{T} \rightarrow \mathbb{T}/G$ the canonical quotient map, is a continuous group homomorphism of $\mathbb{T}$ to $\mathbb{T}$ whose kernel contains G . But by the previous problem, $\theta'(t) = e^{int}$ for some integer n . Therefore $\ker(\theta')$ is finite and hence G is finite.

1.10.12. If $\{k_n\}$ is a sequence of good kernels, show that for $1 \leq p < \infty$, $\|k_n * f - f\|_p \rightarrow 0$ as $n \rightarrow \infty$.

Solution. Suppose that the conclusion is true for any $g \in C(\mathbb{T}) \subseteq L^p(\mathbb{T})$. Let $f \in L^p(\mathbb{T})$. Since $\{k_n\}$ forms a good kernels, $\frac{1}{2\pi} \int_0^\pi |k_n(t)| dt = c < \infty$. As $C(\mathbb{T})$ is dense in $L^p(\mathbb{T})$, given $\epsilon > 0$, there exists $g \in C(\mathbb{T})$ such that $\|f - g\|_p < \frac{\epsilon}{3c}$. By assumption, there exists an $n_0 \in \mathbb{N}$ such that for all $n \geq n_0$ $\|k_n * g - g\|_p < \frac{\epsilon}{3}$. Thus,

$$\begin{aligned} \|k_n * f - f\|_p &\leq \|k_n * f - k_n * g\|_p + \|k_n * g - g\|_p + \|g - f\|_p \\ &= \|k_n * (f - g)\|_p + \|k_n * g - g\|_p + \|g - f\|_p \\ &\leq c\|(f - g)\|_p + \|k_n * g - g\|_p + \|g - f\|_p \\ &< \frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{3} \end{aligned}$$

Thus it is enough to prove for functions in $C(\mathbb{T})$. Let $f \in C(\mathbb{T})$. Then

$$\begin{aligned} \|k_n * f - f\|_p &= \frac{1}{2\pi} \left[\int_0^{2\pi} |k_n * f(x) - f(x)|^p dx \right]^{\frac{1}{p}} \\ &= \frac{1}{2\pi} \left[\int_0^{2\pi} \left| \frac{1}{2\pi} \int_0^{2\pi} k_n(t) f(x-t) dt - f(x) \right|^p dx \right]^{\frac{1}{p}} \\ &= \frac{1}{2\pi} \left[\int_0^{2\pi} \left| \frac{1}{2\pi} \int_0^{2\pi} k_n(t) f(x-t) dt - \frac{1}{2\pi} \int_0^{2\pi} k_n(t) f(x) dt \right|^p dx \right]^{\frac{1}{p}} \\ &\leq \frac{1}{2\pi} \left[\int_0^{2\pi} \frac{1}{2\pi} \int_0^{2\pi} |k_n(t) [f(x-t) - f(x)]|^p dx \right]^{\frac{1}{p}} dt \\ &= \frac{1}{2\pi} \int_0^{2\pi} |k_n(t)| \|f_t - f\|_p dt \\ &= \frac{1}{2\pi} \int_{-\delta}^{2\pi-\delta} |k_n(t)| \|f_t - f\|_p dt \end{aligned}$$

$$= \frac{1}{2\pi} \int_{-\delta}^{\delta} |k_n(t)| \|f_t - f\|_p dt + \int_{\delta}^{2\pi-\delta} |k_n(t)| \|f_t - f\|_p dt,$$

using Minkowski's inequality. Now, given $\epsilon > 0$ there exists $\delta > 0$ such that $\|f_t - f\|_p < \epsilon$, whenever $|t| < \delta$. Now, for this δ there exists an $n_0 \in \mathbb{N}$ such that for all $n \geq n_0$, $\frac{1}{2\pi} \int_{\delta}^{2\pi-\delta} |k_n(t)| dt < \epsilon$. Thus

$$\begin{aligned} \|k_n * f - f\|_p &\leq \frac{1}{2\pi} \left[\epsilon \int_{-\delta}^{\delta} |k_n(t)| dt + 2\|f\|_{\infty} \int_{\delta}^{2\pi-\delta} |k_n(t)| dt \right] \\ &< \epsilon \int_0^{2\pi} |k_n(t)| dt + 2\|f\|_{\infty} \epsilon \\ &= c\epsilon. \end{aligned}$$

1.10.13. If f is of bounded variation on $\mathbb{T}$, then show that $\widehat{f}(n) = O\left(\frac{1}{|n|}\right)$.

Solution. We have

$$\begin{aligned} |\widehat{f}(n)| &= \frac{1}{2\pi} \left| \int_0^{2\pi} e^{-int} f(t) dt \right| \\ &= \frac{1}{2\pi} \left| \int_0^{2\pi} f(t) d\left(\frac{e^{-int}}{-in}\right) \right| \\ &= \frac{1}{2\pi} \left| f(t) \frac{e^{-int}}{-in} \Big|_0^{2\pi} - \int_0^{2\pi} \frac{e^{-int}}{-in} df(t) \right| \\ &= \frac{1}{2\pi|n|} \left| \int_0^{2\pi} e^{-int} df(t) \right| \leq \frac{\text{Var}(f)}{|n|}. \end{aligned}$$

1.10.14. *If the conjugate of Dirichlet kernel is given by*

$$\tilde{D}_N(x) = \sum_{|n| \leq N} \operatorname{sgn}(n) e^{inx},$$

where

$$\operatorname{sgn}(n) = \begin{cases} 1 & n > 0 \\ 0 & n = 0 \\ -1 & n < 0 \end{cases}$$

show that $\int_{-\pi}^{\pi} |\tilde{D}_N(x)| dx \leq c \ln N$.

Solution. We know that

$$\tilde{D}_N(x) = i \frac{\cos(\frac{x}{2}) - \cos(N + \frac{1}{2})x}{\sin(\frac{x}{2})}.$$

Then

$$\begin{aligned} \int_{-\pi}^{\pi} |\tilde{D}_N(x)| dx &= \int_{-\pi}^{\pi} \left| \frac{\cos(\frac{x}{2}) - \cos(N + \frac{1}{2})x}{\sin(\frac{x}{2})} \right| dx \\ &= \int_{-\pi}^{\pi} \left| \frac{\cos(\frac{x}{2}) - \cos Nx \cos(\frac{x}{2}) + \sin Nx \sin(\frac{x}{2})}{\sin(\frac{x}{2})} \right| dx \\ &= \int_{-\pi}^{\pi} \left| \frac{\cos(\frac{x}{2})[1 - \cos(Nx)]}{\sin(\frac{x}{2})} + \sin Nx \right| dx \\ &\leq \int_{-\pi}^{\pi} \left| \frac{1 - \cos(Nx)}{\sin(\frac{x}{2})} \right| dx + 2\pi \\ &= 2 \int_{-\pi}^{\pi} \left| \frac{\sin^2(\frac{Nx}{2})}{\sin(\frac{x}{2})} \right| dx + 2\pi \end{aligned}$$

$$\begin{aligned}
&= 4 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left| \frac{\sin^2 x}{\sin x} \right| dx + 2\pi \\
&= 8 \int_0^{\frac{\pi}{2}} \left| \frac{\sin^2(Nx)}{\sin x} \right| dx + 2\pi \\
&\leq 8 \int_0^{\frac{\pi}{2}} \left| \frac{\sin^2 Nx}{\frac{2x}{\pi}} \right| dx + 2\pi \\
&= 4\pi \int_0^{\frac{\pi}{2}} \left| \frac{\sin^2 Nx}{x} \right| dx + 2\pi \\
&= 4\pi \int_0^{\frac{N\pi}{2}} \frac{|\sin^2 y|}{|y|} dy + 2\pi \\
&= 4\pi \int_0^{\frac{N\pi}{2}} \frac{|\sin y|}{|y|} dy + 2\pi \\
&= 4\pi \left[\int_0^{\frac{\pi}{2}} \left| \frac{\sin x}{x} \right| dx + \sum_{n=2}^N \int_{(n-1)\frac{\pi}{2}}^{\frac{n\pi}{2}} \frac{1}{x} dx + 2\pi \right] \\
&= c + \sum_{n=2}^N \ln x \Big|_{(n-1)\frac{\pi}{2}}^{\frac{n\pi}{2}} + 2\pi \\
&= c + \sum_{n=2}^N \ln \frac{n\frac{\pi}{2}}{(n-1)\frac{\pi}{2}} + 2\pi \\
&= c + \sum_{n=2}^N \ln \left(\frac{n}{n-1} \right) \\
&= c + \ln \left(\prod_{n=2}^N \frac{n}{n-1} \right)
\end{aligned}$$

$$\begin{aligned}
&= c + \ln N \\
&\leq c \ln N.
\end{aligned}$$

1.10.15. Let $f \in C(\pi)$. For each pair of integers m, n , where $0 \leq m < n$, define $\sigma_{m,n}(f, x) = \frac{1}{n-m} \sum_{i=m+1}^n S_i(f, x)$, where $S_i(f, x)$ stands for i -th partial sum of f at x .

(a) Show that $\sigma_{kn, (k+1)n}(f; x) \rightarrow f(x)$ uniformly as $n \rightarrow \infty$, where k, n are integers.

(b) If k, m, n are positive integers with $kn \leq m \leq (k+1)n$ show that $|\sigma_{kn, (k+1)n}(f; x) - S_m(f; x)| \leq \frac{2A}{k}$, where f is such that $|\widehat{f}(j)| \leq \frac{A}{|j|}$ for $j \neq 0$.

(c) Using (a) and (b) show that $S_n(f; x) \rightarrow f(x)$ uniformly on $\mathbb{T}$ whenever $\widehat{f}(n) = O(\frac{1}{n})$.

Solution. For integers m and n , we have $\sigma_{m,n}(f; x) = \frac{1}{n-m} \sum_{i=m+1}^n S_i(f; x)$. Rewriting this in terms of the kernel, we obtain

$$\sigma_{m,n}(f; x) = \frac{1}{n-m} [(n+1)\sigma_{n+1}(f; x) - (m+1)\sigma_{m+1}(f; x)] \quad (1.17)$$

Using the Fourier expansion of the kernel, one obtains the following Fourier expansion for $\sigma_{m,n}(f)$ as follows:

$$\sigma_{m,n}(f; x) = S_m(f; x) + \sum_{m < |j| \leq n} \frac{n+1-|j|}{n-m} \widehat{f}(j) e^{ijx}. \quad (1.18)$$

By (1.17), we have

$$\begin{aligned}
\sigma_{kn, (k+1)n}(f; x) &= \frac{1}{n} [(k+1)n+1)\sigma_{(k+1)n}(f; x) - (kn+1)\sigma_{kn+1}(f; x)] \\
&= (k+1 + \frac{1}{n})\sigma_{(k+1)n}(f; x) - (k + \frac{1}{n})\sigma_{kn+1}(f; x)
\end{aligned} \quad (1.19)$$

which converges uniformly to $(k+1)f(x) - kf(x) = f(x)$ as $n \rightarrow \infty$, by using problem 1.10.12. This proves (a).

To prove (b), we use (1.18). In fact, from (1.18), we get

$$|\sigma_{kn, (k+1)n}(f; x) - S_m(f; x)| \leq \sum_{|kn| \leq |j| \leq (k+1)n} |\hat{f}(j)| \leq 2 \sum_{j=kn+1}^{(k+1)n} \frac{A}{j} \leq \frac{2A}{k}.$$

For (c), given that $\hat{f}(n) = O(\frac{1}{n})$, that is $|\hat{f}(n)| \leq \frac{A}{|n|}$, for $n \neq 0$. Let $\epsilon > 0$ be given.

Choose an integer $k > 0$ such that $\frac{A}{k} < \frac{\epsilon}{4}$. By (a) we can find $n_0 \geq k$ such that for all $n \geq n_0$, $|\sigma_{kn, (k+1)n}(f; x) - f(x)| < \frac{\epsilon}{2}$. Let $m \geq kn_0$ such that for some $n \geq n_0$, $kn \leq m < (k+1)n$. By (b), we have $|\sigma_{kn, (k+1)n}(f; x) - S_m(f; x)| < \frac{2A}{k} < \frac{\epsilon}{2}$. Thus by the above two inequalities, (c) follows.

1.10.16. Show that $\sum_{n=1}^{\infty} (-1)^{n-1}$ is Cesaro summable to $\frac{1}{2}$.

Solution. $\sum_{n=1}^{\infty} (-1)^{n-1}$. Here $a_n = \begin{cases} 1 & \text{if } n \text{ is odd} \\ -1 & \text{if } n \text{ is even.} \end{cases}$

Here s_n is the n -th partial sum of a_n and $s_n = \begin{cases} 1 & \text{if } n \text{ is odd} \\ 0 & \text{if } n \text{ is even.} \end{cases}$

Consider

$$\begin{aligned} \sigma_n &= \frac{s_1 + s_2 + \cdots + s_n}{n}. \\ \sigma_{2n} &= \frac{s_1 + s_2 + \cdots + s_{2n}}{2n} \\ &= \frac{n}{2n} \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{2} \quad \forall n \in \mathbb{N}. \\
\lim_{n \rightarrow \infty} \sigma_{2n} &= \frac{1}{2} \\
\sigma_{2n-1} &= \frac{s_1 + s_2 + \cdots + s_{2n-1}}{2n-1} \\
&= \frac{n-1+1}{2n-1} = \frac{n}{2n-1}.
\end{aligned}$$

Therefore $\sigma_{2n-1} \rightarrow \frac{1}{2}$ as $n \rightarrow \infty$

$$\lim_{n \rightarrow \infty} \sigma_n = \frac{1}{2}.$$

Hence $\sum_{n=1}^{\infty} (-1)^{n-1}$ is Cesaro summable to $\frac{1}{2}$.

1.10.17. Show that $\sum_{n=0}^{\infty} (-1)^n (n+1)$ is not Cesaro summable but Able summable to $\frac{1}{4}$.

Solution. We know that $\sum_{n=0}^{\infty} (-1)^n (n+1) r^n = 1 - 2r + 3r^2 - \cdots = (1+r)^{-2}$ for $0 < r < 1$. Then

$$\lim_{r \rightarrow 1} \sum_{n=0}^{\infty} (-1)^n (n+1) r^n = \lim_{r \rightarrow 1} (1+r)^{-2} = \frac{1}{4}.$$

Therefore, $\sum_{n=0}^{\infty} (-1)^n (n+1)$ is Abel summable to $\frac{1}{4}$. Now consider the series

$$\sum_{n=0}^{\infty} (-1)^n (n+1) = \sum_{n=1}^{\infty} (-1)^{n-1} n.$$

$$\text{Here } s_n = \begin{cases} \frac{-n}{2} & \text{if } n \text{ is even} \\ \frac{n+1}{2} & \text{if } n \text{ is odd.} \end{cases}$$

Clearly,

$$\sigma_{2n} = \frac{s_1 + s_2 + \cdots + s_{2n}}{2n} = 0 \quad \text{for all } n,$$

$$\begin{aligned}
\sigma_{2n-1} &= \frac{s_1 + s_2 + \cdots + s_{2n-1}}{2n-1} \\
&= \frac{2n-1+1}{2(2n-1)} \\
&= \frac{n}{2n-1}.
\end{aligned}$$

Therefore, $\sigma_{2n-1} \rightarrow \frac{1}{2}$ as $n \rightarrow \infty$. Hence $\sum_{n=0}^{\infty} (-1)^n (n+1)$ is not Cesaro summable.

1.10.18. Prove that if $\sum_{i=1}^{\infty} c_i$ converges to s , then $\sum_{i=1}^{\infty} c_i$ is Cesaro summable to s .

Solution. Let $s_n = \sum_{i=1}^n c_i$. The assumption is that $s_n \rightarrow s$ as $n \rightarrow \infty$. Given $\epsilon > 0$ there exists $m \in \mathbb{N}$ such that $|s_n - s| < \epsilon$ for all $n > m$. Now for all $n > m$, we can write

$$\begin{aligned}
|\sigma_n - s| &= \left| \frac{s_1 + s_2 + \cdots + s_n}{n} - s \right| \\
&= \left| \frac{(s_1 - s) + \cdots + (s_m - s) + (s_{m+1} - s) + \cdots + (s_n - s)}{n} \right| \\
&\leq \frac{\sum_{i=1}^m (s_i - s)}{n} + \frac{\sum_{i=m+1}^n (s_i - s)}{n} \\
&\leq \frac{A}{n} + \frac{n-m}{n} \epsilon \\
&\leq \frac{A}{n} + \epsilon,
\end{aligned}$$

where $A = \sum_{i=1}^m (s_i - s)$. Now we can find $m_1 \in \mathbb{N}$ such that $\frac{A}{n} < \epsilon$ for all $n \geq m_1$.

Choose $m_0 = \max\{m, m_1\}$. Then $|\sigma_n - s| < \epsilon$ for all $n \geq m_0$. The series $\sum_{n=1}^{\infty} c_n$ is Cesaro summable to s .

1.10.19. Assume that $\sum_{n=1}^{\infty} a_n$ is Cesaro summable to l and $\lim_{n \rightarrow \infty} na_n = 0$. Then show that $\sum_{n=1}^{\infty} a_n$ converges to l .

Solution. Assume that $\sum_{n=1}^{\infty} a_n$ is Cesaro summable to l and $\lim_{n \rightarrow \infty} na_n = 0$. Consider,

$$s_n = \sum_{i=1}^n a_i \text{ and } \sigma_n = \frac{s_1 + s_2 + \cdots + s_n}{n}.$$

By the assumption, $\lim_{n \rightarrow \infty} \sigma_n = l$

$$\begin{aligned} \sigma_n &= \frac{s_1 + s_2 + \cdots + s_n}{n} \\ &= \frac{a_1 + (a_1 + a_2) + \cdots + (a_1 + a_2 + \cdots + a_n)}{n} \\ &= \sum_{i=1}^n \left(1 - \frac{i-1}{n}\right) a_i. \end{aligned}$$

Now

$$\begin{aligned} \sigma_n - s_n &= \sum_{i=1}^n \left(1 - \frac{i-1}{n}\right) a_i - \sum_{i=1}^n a_i \\ &= \frac{1}{n} \sum_{i=1}^n a_i - \frac{1}{n} \sum_{i=1}^n i a_i \\ &= \frac{s_n}{n} - \frac{1}{n} \sum_{i=1}^n i a_i \\ \left(1 + \frac{1}{n}\right) s_n &= \sigma_n + \frac{1}{n} \sum_{i=1}^n i a_i \end{aligned} \tag{1.20}$$

Since $\lim_{n \rightarrow \infty} na_n = 0$, $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n i a_i = 0$. Taking limit as $n \rightarrow \infty$ in 1.20, we have $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right) s_n = \lim_{n \rightarrow \infty} \sigma_n = l$. Thus $\lim_{n \rightarrow \infty} s_n = l$.

Here we have used the following result. If $\lim_{n \rightarrow \infty} a_n = m \neq 0$, $\lim_{n \rightarrow \infty} a_n b_n = l$ then $\lim_{n \rightarrow \infty} b_n$ exists and equals $\frac{l}{m}$.

1.10.20. If $\sum_{n=1}^{\infty} C_n$ is Cesaro summable to l , then prove that $\sum_{n=1}^{\infty} C_n$ is Abel summable to l .

Solution. Assume that $\sum_{n=1}^{\infty} C_n$ is Cesaro summable to l . Define s_n and σ_n as before then $\lim_{n \rightarrow \infty} \sigma_n = l$. It can be easily seen that $C_1 = s_1$; $C_n = s_n - s_{n-1}$ for all $n \geq 2$ and $s_1 = \sigma_1$, $s_n = n\sigma_n - (n-1)\sigma_{n-1}$ for all $n \geq 2$. Consider $0 < r < 1$. Now it can be easily seen that $\sum_{n=1}^{\infty} n\sigma_n r^{n-1}$ converges absolutely for all $0 \leq r < 1$. Then

$$\begin{aligned}
(1-r)^2 \sum_{n=1}^{\infty} n\sigma_n r^{n-1} &= (1-r) \sum_{n=1}^{\infty} [n\sigma_n r^{n-1} - n\sigma_n r^n] \\
&= (1-r) \left[\sum_{n=1}^{\infty} n\sigma_n r^{n-1} - \sum_{n=1}^{\infty} n\sigma_n r^n \right] \\
&= (1-r) \left[\sigma_1 + \sum_{n=2}^{\infty} n\sigma_n r^{n-1} - \sum_{n=2}^{\infty} (n-1)\sigma_{n-1} r^{n-1} \right] \\
&= (1-r) \left[s_1 + \sum_{n=2}^{\infty} (n\sigma_n - (n-1)\sigma_{n-1}) r^{n-1} \right] \\
&= (1-r) \left[s_1 + \sum_{n=2}^{\infty} s_n r^{n-1} \right] \\
&= (1-r) \sum_{n=1}^{\infty} s_n r^{n-1} \\
&= \sum_{n=1}^{\infty} s_n r^{n-1} - \sum_{n=1}^{\infty} s_n r^n \\
&= s_1 + \sum_{n=2}^{\infty} s_n r^{n-1} - \sum_{n=2}^{\infty} s_{n-1} r^{n-1} \\
&= s_1 + \sum_{n=2}^{\infty} (s_n - s_{n-1}) r^{n-1} \\
&= c_1 + \sum_{n=2}^{\infty} C_n r^{n-1} = \sum_{n=1}^{\infty} C_n r^{n-1}.
\end{aligned}$$

Since $\sum_{n=1}^{\infty} n\sigma_n r^{n-1}$ converges for $0 \leq r < 1$, it follows that $\sum_{n=1}^{\infty} C_n r^{n-1}$ converges for $0 \leq r < 1$. Define $f(r) = \sum_{n=1}^{\infty} C_n r^{n-1}$ for all $0 \leq r < 1$. Now we shall show that $\lim_{r \rightarrow 1} f(r) = l$. Choose $\epsilon > 0$. Then there exists $m \in \mathbb{N}$ such that $|\sigma_n - l| < \epsilon$ for all $n \geq m$. Now choose $\delta = \epsilon$. Then for all $1 - \epsilon < r < 1$, we have

$$\begin{aligned}
|f(r) - l| &= \left| \sum_{n=1}^{\infty} C_n r^{n-1} - l \right| \\
&= \left| (1-r)^2 \sum_{n=1}^{\infty} n\sigma_n r^{n-1} - l(1-r)^2 \sum_{n=1}^{\infty} n r^{n-1} \right| \\
&= (1-r)^2 \left| \sum_{n=1}^{\infty} n(\sigma_n - l) r^{n-1} \right| \\
&\leq (1-r)^2 \sum_{n=1}^{\infty} n |\sigma_n - l| r^{n-1} \\
&= (1-r)^2 \left[\sum_{n=1}^{m-1} n r^{n-1} |\sigma_n - l| + \epsilon \sum_{n=m}^{\infty} n r^{n-1} \right] \\
&= (1-r)^2 \left[\sum_{n=1}^{m-1} n |\sigma_n - l| + \epsilon (1-r)^{-2} \right] \\
&= k(1-r)^2 + \epsilon, \text{ where } k = \sum_{n=1}^{m-1} n |\sigma_n - l| \\
&\leq k\epsilon^2 + \epsilon.
\end{aligned}$$

Now for any $\epsilon' > 0$, there exists unique $\epsilon > 0$ (in fact other root is negative) such that $k\epsilon^2 + \epsilon = \epsilon'$. Hence $\lim_{r \rightarrow 1} f(r) = l$.

1.10.21. Assume that f is periodic and monotone on $[-\pi, \pi]$. Then show that $\widehat{f}(n) = O\left(\frac{1}{|n|}\right)$.

Solution. Assume that f is monotone. Then by Lebesgue-Young theorem f is differentiable a.e and $f' \in L^1[-\pi, \pi]$. Now

$$\begin{aligned}\widehat{f}(n) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) e^{-int} dt \\ &= \frac{1}{2\pi} \left[f(t) \frac{e^{-int}}{-in} \Big|_{-\pi}^{\pi} + \int_{-\pi}^{\pi} f'(t) \frac{e^{-int}}{in} dt \right].\end{aligned}\tag{1.21}$$

The first term of the R.H.S of (1.21) is zero since $f(\pi) = f(-\pi)$. Then $\widehat{f}(n) = \frac{1}{2\pi in} \int_{-\pi}^{\pi} f'(t) e^{-int} dt$ which implies

$$\begin{aligned}|\widehat{f}(n)| &\leq \frac{1}{2\pi|n|} \int_{-\pi}^{\pi} |f'(t)| dt \\ &= \frac{K}{2\pi|n|}, \quad \text{since } f' \in L^1[-\pi, \pi] \\ \widehat{f}(n) &= O\left(\frac{1}{|n|}\right).\end{aligned}$$

1.11 Exercises

1.11.1. Let $f \in L^1(\mathbb{T})$. Show that $\lim_{\delta \rightarrow 0} \int_{-\pi}^{\pi} |f(x + \delta) - f(x)| dx = 0$.

1.11.2. Show that Fourier series of the function

$$f(x) = \begin{cases} 1 & x \text{ is rational} \\ 0 & x \text{ is irrational} \end{cases}$$

is identically 0 and does not converge to f for any rational number x .